Introduction to stochastic process (STAT3007)

Chapter 2. Stochastic processes

Department of Statistics, CUHK

随机过程导论 (STAT3007)

第2章. 随机过程

统计系，港大

Brief Introduction

A stochastic process/random process is a sequence of random variable is a collection of random variables $\{X_t\}_{t\in\mathcal{T}}$. indexed by a set $\mathcal{T}$,

- the index, often representing time, can be discrete ($\mathcal{T} = \{0, 1, 2, \cdots\}$), or continuous ($\mathcal{T} = [0, \infty)$);
- the goal for this chapter is to study some well-studied stochastic processes:
  - *understand the definition:* establish the rigorous mathematical definition and understand the intuition behind it;
  - *characterize the distributions:* identify the probability distributions associated with the process;
  - *explore key properties:* study the mathematical and probabilistic properties relevant to understanding and working with the process.

随机过程/随机过程是一个随机变量的序列
变量是一个随机变量集合 $\{X_t\}_{t\in\mathcal{T}}$，由一个集合 $\mathcal{T}$ 索引，

索引，通常表示时间，可以是离散的 ($\mathcal{T} = \{0, 1, 2, \cdots\}$)，或连续的 ($\mathcal{T} = [0, \infty)$)；
本章的目标是研究一些经过充分研究的随机过程：

理解定义：建立严谨的数学定义并理解其背后的直观概念；

表征分布：识别概率
与该过程相关的分布；
探索关键性质：研究数学和
与理解和处理该过程相关的概率性质。

Outline

1. Bernoulli process
   - Definition: counting success
   - Distributions associated with Bernoulli process
   - Law of large number
   - **Convergence of random variables*
2. Poisson process
   - Continuous arrivals and definition
   - Distributions associated with Poisson process
   - Superposition and thinning
3. Simple random walk and Brownian motion
   - Simple random walk and continuous limit
   - Brownian motion
   - Central limit theorem

大纲

1. 伯努利过程
   - 定义：计数成功
   - 与伯努利过程相关的分布
   - 大数定律
   - *随机变量收敛
2. 泊松过程
   - 连续到达和定义
   - 与泊松过程相关的分布
   - 叠加和稀疏
3. 简单随机游走和布朗运动
   - 简单随机游走和连续极限
   - 布朗运动
   - 中心极限定理

Outline

1. Bernoulli process
   - Definition: counting success
   - Distributions associated with Bernoulli process
   - Law of large number
   - **Convergence of random variables*
2. Poisson process
   - Continuous arrivals and definition
   - Distributions associated with Poisson process
   - Superposition and thinning
3. Simple random walk and Brownian motion
   - Simple random walk and continuous limit
   - Brownian motion
   - Central limit theorem

概要

1. 伯努利过程
   - 定义：计数成功
   - 与伯努利过程相关的分布
   - 大数定律
   - *随机变量收敛
2. 泊松过程
   - 连续到达和定义
   - 与泊松过程相关的分布
   - 叠加和稀疏
3. 简单随机游走和布朗运动
   - 简单随机游走和连续极限
   - 布朗运动
   - 中心极限定理

Outline

1. Bernoulli process
   - Definition: counting success
   - Distributions associated with Bernoulli process
   - Law of large number
   - **Convergence of random variables*
2. Poisson process
   - Continuous arrivals and definition
   - Distributions associated with Poisson process
   - Superposition and thinning
3. Simple random walk and Brownian motion
   - Simple random walk and continuous limit
   - Brownian motion
   - Central limit theorem

概要

1. 伯努利过程
   - 定义：计数成功
   - 与伯努利过程相关的分布
   - 大数定律
   - *随机变量收敛
2. 泊松过程
   - 连续到达和定义
   - 与泊松过程相关的分布
   - 叠加和稀疏
3. 简单随机游走和布朗运动
   - 简单随机游走和连续极限
   - 布朗运动
   - 中心极限定理

Bernoulli distribution and Bernoulli process

We start this chapter by one of *the simplest kind of random process*: a sequence of independent trials where success or failure occur with given probabilities.

- **Bernoulli distribution:** $X \sim \text{Bern}(p)$ represents a single binary experiment with success probability p.
  - pmf: $\mathbb{P}(X=1)=p$, $\mathbb{P}(X=0)=1-p$;
  - $\mathbb{E}[X]=p$, $\text{Var}(X)=p(1-p)$, $M_X(t)=pe^t+1-p$.
- **Bernoulli process:** a sequence $\{X_1, X_2, \cdots\}$ of independent Bernoulli variables with $X_i \sim \text{Bern}(p)$;
  - independence between $\{X_i\}$ uniquely determine the joint distribution of $(X_{i_1}, \cdots, X_{i_k})$ for any indices $i_1, \cdots, i_k$, which in turn uniquely determines the distribution of the process (*A rigorous proof needs Kolmogorov's extension theorem*).

本章从一种最简单的随机过程开始：一系列独立的试验，其中成功或失败以给定的概率发生。

- 伯努利分布： $X \sim \text{Bern}(p)$ 表示一个单一的二元实验，成功概率为 p。
  - 概率质量函数：$\mathbb{P}(X=1)=p$, $\mathbb{P}(X=0)=1-p$;
  - $\mathbb{E}[X]=p$, $\text{Var}(X)=p(1-p)$, $M_X(t)=pe^t+1-p$.
- 伯努利过程：一个由 $\{X_1, X_2, \cdots\}$ 个独立伯努利变量组成的序列 $X_i \sim \text{Bern}(p)$;
  - $\{X_i\}$ 之间的独立性唯一地确定了 $(X_{i_1}, \cdots, X_{i_k})$ 的联合分布，对于任何索引 i_1 $\cdots$ i_k, , , 这反过来又唯一地确定了该过程的分布（严格的证明需要柯尔莫哥洛夫延拓定理）。

Key terms associated with the Bernoulli process

Even in this very simple model, there are many natural questions we could try to answer. For example:

- How many successes do we have by time n?
- How long must we wait for the first success? Between successes?

These are closely related to the following quantities determined by the Bernoulli process

- number of successes by time n: $S_n = X_1 + X_2 + \cdots + X_n$
- k-th time of success: $T_k = \min\{n : S_n = k\}$.

即使在这个非常简单的模型中，我们也可以尝试回答许多自然的问题。例如：

到时间 n 我们有多少次成功?

我们必须等待多久才能获得第一次成功？两次成功之间?

这些与以下由伯努利过程确定的量密切相关

时间内的成功次数 n: $S_n = X_1 + X_2 + \cdots + X_n$

k-次成功的时间： $T_k = \min\{n : S_n = k\}$。

Outline

概要

Binomial distribution: number of successes at time n

The **Binomial distribution** represents the number of successes in a Bernoulli process, that is $S_n \sim \mathsf{B}(n,p)$:

- pmf: it can be written in terms of the binomial coefficients (also called combination number) C_n^k:

$$\mathbb{P}(S_n = k) = C_n^k p^k (1-p)^{n-k}, \text{ for } k = 0,1,2,\cdots,n.$$

  We sometimes use $\binom{n}{k}$ or C_k^n to represent C_n^k.

- $\mathbb{E}[S_n] = np$, $\mathsf{Var}(S_n) = np(1-p)$ and $M_{S_n}(t) = (pe^t + 1 - p)^n$,
  - can be derived using the Binomial theorem:

$$(x+y)^n = \sum_{k=0}^{n} C_n^k x^k y^{n-k};$$

  - a more elegant approach is to make use of the independent sum by

$$S_n = X_1 + X_2 + \cdots + X_n.$$

二项分布表示伯努利过程中的成功次数，即 $S_n \sim \mathsf{B}(n,p)$:

概率质量函数：它可以表示为二项系数（也称为组合数） C_n^k:

$$\mathbb{P}(S_n = k) = C_n^k p^k (1-p)^{n-k}, \text{ for } k = 0,1,2,\cdots,n.$$

我们有时使用 $\binom{n}{k}$ 或 C_k^n 来表示 C_n^k。

$\mathbb{E}[S_n] = np$, $\mathsf{Var}(S_n) = np(1-p)$ and $M_{S_n}(t) = (pe^t + 1 - p)^n$,

可以用二项式定理推导出来:

$$(x+y)^n = \sum_{k=0}^{n} C_n^k x^k y^{n-k};$$

一种更优雅的方法是利用独立和

$$S_n = X_1 + X_2 + \cdots + X_n.$$

Geometric distribution: first success time

The **Geometric distribution** represents the first success time in a Bernoulli process, that is $T_1 \sim \text{Geo}(p)$:

- pmf: it can be derived from the structure of the Bernoulli process

$$\mathbb{P}(T_1 = n) = (1-p)^{n-1}p, \text{ for } n = 0, 1, 2, \cdots$$

(In alternative definitions, the Geometric distribution counts the number of failures before the first success.)

- $\mathbb{E}[T_1] = \frac{1}{p}$, $\text{Var}(T_1) = \frac{1-p}{p^2}$ and $M_{T_1}(t) = \frac{pt}{1-(1-p)e^t}$,
  - follows from the sum of geometric series:

$$\sum_{i=0}^{\infty} ar^i = \frac{a}{1-r}, \text{ for } |r| < 1.$$

几何分布表示伯努利过程中的首次成功时间，即 $T_1 \sim \text{Geo}(p)$:

概率质量函数：它可以从伯努利过程的结构推导出来

$$\mathbb{P}(T_1 = n) = (1-p)^{n-1}p, \text{ for } n = 0, 1, 2, \cdots$$

（在替代定义中　　　，几何分布计算的是首次成功之前的失败次数。）

$\mathbb{E}[T_1] = \frac{1}{p}$, $\text{Var}(T_1) = \frac{1-p}{p^2}$ and $M_{T_1}(t) = \frac{pt}{1-(1-p)e^t}$,

由几何级数之和推导而来：

$$\sum_{i=0}^{\infty} ar^i = \frac{a}{1-r}, \text{ for } |r| < 1.$$

Negative binomial distribution: r-th success time

The **Negative binomial distribution**(also known as Pascal distribution) represents the r-th success time, that is $T_r \sim \text{NB}(r, p)$:

- pmf: it can be derived from the structure of the Bernoulli process

$$\mathbb{P}(T_r = n) = C_{n-1}^{r-1}(1-p)^{n-r}p^r, \text{ for } n = 0, 1, 2, \cdots$$

- $\mathbb{E}[T_r] = \frac{r}{p}$, $\text{Var}(T_r) = \frac{r(1-p)}{p^2}$ and $M_{T_r}(t) = \left(\frac{pt}{1-(1-p)e^t}\right)^r$,
  - can be obtained by applying the Generalized Binomial theorem:

$$(x+y)^{-n} = \sum_{k=0}^{\infty} C_{n+k-1}^{k}(-x)^{-k}y^{-n-k};$$

  - there is another way to calculate these quantities by observing the that: *waiting times between successes are independent and identically distributed (i.i.d.).*

负二项分布（也称为帕斯卡分布）表示第 r 次成功的时间，即 $T_r \sim \text{NB}(r, p)$:

概率质量函数：它可以从伯努利过程的结构推导出来

$$\mathbb{P}(T_r = n) = C_{n-1}^{r-1}(1-p)^{n-r}p^r, \text{ for } n = 0, 1, 2, \cdots$$

$\mathbb{E}[T_r] = \frac{r}{p}$, $\text{Var}(T_r) = \frac{r(1-p)}{p^2}$ and $M_{T_r}(t) = \left(\frac{pt}{1-(1-p)e^t}\right)^r$,

可以通过应用广义二项式定理来获得:

$$(x+y)^{-n} = \sum_{k=0}^{\infty} C_{n+k-1}^{k}(-x)^{-k}y^{-n-k};$$

还有另一种方法可以通过观察来计算这些量：成功之间的等待时间是独立同分布的（*i.i.d.*）。

i.i.d. waiting times between successes

Let us consider the distribution of $T_1, T_2 - T_1, \cdots, T_k - T_{k-1}$:

- for $n_1, \cdots, n_k$, we are able to calculate that

$$\begin{aligned}&\mathbb{P}(T_1 = n_1, T_2 - T_1 = n_2, \cdots, T_k - T_{k-1} = n_k)\\&\quad = (1-p)^{n_1-1} \cdot p \cdot (1-p)^{n_2-1} \cdot p \cdots (1-p)^{n_k-1} \cdot p;\end{aligned}$$

- we know that $\mathbb{P}(T_1 = n) = (1-p)^{n-1}p$;
- we can also prove that $\mathbb{P}(T_j - T_{j-1} = n) = (1-p)^{n-1}p$;
- this concludes that

$$\begin{aligned}&\mathbb{P}(T_1 = n_1, T_2 - T_1 = n_2, \cdots, T_k - T_{k-1} = n_k)\\&\quad = \mathbb{P}(T_1 = n_1)\mathbb{P}(T_2 - T_1 = n_2) \cdots \mathbb{P}(T_k - T_{k-1} = n_k),\end{aligned}$$

  which proves the desired i.i.d. property;
- such property can be interpreted as *memoryless property*, meaning that the waiting time until the next success is independent of how many trials have already occurred.

让我们考虑 T_1 $T_2 - T_1$ $\cdots$ $T_k - T_{k-1}$:的分布，，

对于 $n_1, \cdots, n_k$，我们能够计算出

$$\begin{aligned}&\mathbb{P}(T_1 = n_1, T_2 - T_1 = n_2, \cdots, T_k - T_{k-1} = n_k)\\&\quad = (1-p)^{n_1-1} \cdot p \cdot (1-p)^{n_2-1} \cdot p \cdots (1-p)^{n_k-1} \cdot p;\end{aligned}$$

我们知道 $\mathbb{P}(T_1 = n) = (1-p)^{n-1}p$;

我们也可以证明 $\mathbb{P}(T_j - T_{j-1} = n) = (1-p)^{n-1}p$;

这表明

$$\begin{aligned}&\mathbb{P}(T_1 = n_1, T_2 - T_1 = n_2, \cdots, T_k - T_{k-1} = n_k)\\&\quad = \mathbb{P}(T_1 = n_1)\mathbb{P}(T_2 - T_1 = n_2) \cdots \mathbb{P}(T_k - T_{k-1} = n_k),\end{aligned}$$

这证明了所需的独立同分布属性;

这种属性可以解释为无记忆性，意味着下一次成功的等待时间与已经发生的试验次数无关。

Outline

1. Bernoulli process
   - Definition: counting success
   - Distributions associated with Bernoulli process
   - Law of large number
   - **Convergence of random variables*
2. Poisson process
   - Continuous arrivals and definition
   - Distributions associated with Poisson process
   - Superposition and thinning
3. Simple random walk and Brownian motion
   - Simple random walk and continuous limit
   - Brownian motion
   - Central limit theorem

概要

1. 伯努利过程
   - 定义：计数成功
   - 与伯努利过程相关的分布
   - 大数定律
   - *随机变量收敛
2. 泊松过程
   - 连续到达和定义
   - 与泊松过程相关的分布
   - 叠加和稀疏
3. 简单随机游走和布朗运动
   - 简单随机游走和连续极限
   - 布朗运动
   - 中心极限定理

Intuition: frequency and probability

- One of the most fundamental intuitions in probability is the frequency interpretation: *If we repeat the random experiment many times, the probability $\mathbb{P}(A)$ is the fraction of these experiments in which A occurs.*
- Under the modeling of Bernoulli process, we can let X_i representing whether A occurs or not at the i-th trial and then the frequency, or the fraction of these experiments in which A occurs is

$$\frac{1}{n}(X_1 + X_2 + \cdots + X_n) = \frac{S_n}{n}.$$

- This intuition can be formalized mathematically as

$$\lim_{n\to\infty} \frac{S_n}{n} = p,$$

where p is the Bernoulli parameter. *Note: The limit notation here may represent convergence in probability (weak), almost sure convergence (strong), or convergence in distribution.*

概率中最基本的直观之一是频率解释：如果我们重复随机实验很多次，概率 $\mathbb{P}(A)$ 就是这些实验中 A 发生的比例。

在伯努利过程的建模下，我们可以让 X_i 表示在 i-次试验中 A 是否发生，然后频率，即这些实验中 A 发生的比例是

$$\frac{1}{n}(X_1 + X_2 + \cdots + X_n) = \frac{S_n}{n}.$$

这种直观可以用数学形式化表示为

$$\lim_{n\to\infty} \frac{S_n}{n} = p,$$

p 是伯努利参数。注意：这里的极限符号可能表示依概率收敛（弱）、几乎必然收敛（强）或依分布收敛。

Law of large number

A slight extension of the previous principle yields another fundamental intuition—the average interpretation of expectation: *If we repeat the random experiment many times, the expectation $\mathbb{E}[X]$ is the average of the outcomes of X over all these experiments*

- This can be formalized mathematically as follows: Let $X_1, X_2, \cdots$ be i.i.d. random variables with the same distribution as X, then

$$\lim_{n\to\infty} \frac{1}{n}\sum_{i=1}^{n} X_i = \mathbb{E}[X].$$

This result is known as **Law of Large Numbers (LLN)**.

- A quick interpretation through discrete random variable why this can be extended by the previous principle is the following:

$$\begin{aligned}\lim_{n\to\infty} \frac{1}{n}\sum_{i=1}^{n} X_i &= \lim_{n\to\infty} \sum_{i=1}^{n} \frac{1}{n} \sum_{x\in D_X} x \mathbb{1}_{\{X_i=x\}} \\ &= \sum_{x\in D_X} x \cdot \lim_{n\to\infty} \sum_{i=1}^{n} \frac{1}{n} \mathbb{1}_{\{X_i=x\}} = \sum_{x\in D_X} x \cdot \mathbb{P}(X=x) = \mathbb{E}[X].\end{aligned}$$

对前述原理的轻微扩展会引出另一个基本直觉——期望的平均解释：如果我们重复进行随机实验很多次，期望 $\mathbb{E}[X]$ 是所有这些实验中 X 的平均结果。

这可以数学上形式化为如下：设 $X_1, X_2, \cdots$ 是独立同分布的随机变量，且与 X 具有相同的分布，则

$$\lim_{n\to\infty} \frac{1}{n}\sum_{i=1}^{n} X_i = \mathbb{E}[X].$$

这一结果被称为大数定律（LLN）。
对离散随机变量的快速解释，说明为什么这可以由前面的原则扩展，如下：

$$\begin{aligned}\lim_{n\to\infty} \frac{1}{n}\sum_{i=1}^{n} X_i &= \lim_{n\to\infty} \sum_{i=1}^{n} \frac{1}{n} \sum_{x\in D_X} x \mathbb{1}_{\{X_i=x\}} \\ &= \sum_{x\in D_X} x \cdot \lim_{n\to\infty} \sum_{i=1}^{n} \frac{1}{n} \mathbb{1}_{\{X_i=x\}} = \sum_{x\in D_X} x \cdot \mathbb{P}(X=x) = \mathbb{E}[X].\end{aligned}$$

Proof of LLN

We now present a formal proof of LLN:

- The main idea is to show that

$$\mathbb{E}\Big[\lim_{n\to\infty}\frac{S_n}{n}\Big]=\lim_{n\to\infty}\mathbb{E}\Big[\frac{S_n}{n}\Big]=p,\ \text{and}\ \mathsf{Var}\Big[\lim_{n\to\infty}\frac{S_n}{n}\Big]=\lim_{n\to\infty}\mathsf{Var}\Big[\frac{S_n}{n}\Big]=0,$$

  then use the fact that an r.v. with zero variance must be constant.

- The rest is standard,
  - for the mean, we have $\mathbb{E}\big[\frac{S_n}{n}\big]=\frac{1}{n}\cdot n\cdot\mathbb{E}[X_1]=p$;
  - for the variance, we have

$$\mathsf{Var}\Big(\frac{S_n}{n}\Big)=\frac{1}{n^2}\cdot n\cdot\mathsf{Var}[X_1]=\frac{p(1-p)}{n}\to 0\ \text{as}\ n\to\infty.$$

- *A more rigorous proof requires a precise definition of convergence, and in particular, different modes of convergence require different proofs.*

我们现在给出大数定律的正式证明：

主要思想是要说明

$$\mathbb{E}\Big[\lim_{n\to\infty}\frac{S_n}{n}\Big]=\lim_{n\to\infty}\mathbb{E}\Big[\frac{S_n}{n}\Big]=p,\ \text{and}\ \mathsf{Var}\Big[\lim_{n\to\infty}\frac{S_n}{n}\Big]=\lim_{n\to\infty}\mathsf{Var}\Big[\frac{S_n}{n}\Big]=0,$$

然后利用这样一个事实：方差为零的随机变量必须是常数。

其余部分是标准的，

对于均值，我们有 $\mathbb{E}\big[\frac{S_n}{n}\big]=\frac{1}{n}\cdot n\cdot\mathbb{E}[X_1]=p$;

对于方差，我们有

$$\mathsf{Var}\Big(\frac{S_n}{n}\Big)=\frac{1}{n^2}\cdot n\cdot\mathsf{Var}[X_1]=\frac{p(1-p)}{n}\to 0\ \text{as}\ n\to\infty.$$

更严谨的证明需要精确的收敛定义，并且，特别是不同的收敛模式需要不同的证明。

Outline

概要

Convergence of real number

To properly understand convergence of random variables, we first revisit the formal definition of convergence in the real number setting

- Consider a sequence of real number $\{a_n\}$, we call a_n **converges to** a if and only if
  - for any $\epsilon > 0$, there exists an N such that

$$|a_n - a| < \epsilon \text{ for all } n \geq N.$$

- We use the following notations to represent such convergence:
  - $\lim_{n\to\infty} a_n = a$;
  - $a_n \to a$ as $n \to \infty$.
- The key point is: although a random variable produces real numbers as outputs, we must remember that it is fundamentally a *function*. Therefore, convergence of random variables means *convergence of functions*, not just convergence of individual real numbers

实数收敛

要正确理解随机变量收敛，我们首先重新审视实数环境下的收敛形式定义

考虑一个实数序列 $\{a_n\}$，如果且仅当对于任何 $\{a_n\}$，都存在一个 a_n，使得

对于任何 $\epsilon > 0$，都存在一个 N，使得

$$|a_n - a| < \epsilon \text{ for all } n \geq N.$$

我们使用以下符号来表示这种收敛：

$\lim_{n\to\infty} a_n = a$;

$a_n \to a$ as $n \to \infty$.

关键点在于：虽然随机变量以实数作为输出，但我们必须记住它本质上是一个函数。因此，随机变量收敛意味着函数收敛，而不仅仅是单个实数收敛

Major Types of Convergence for Random Variables

Consider a sequence of random variables $\{X_n\}_{n\in\mathbb{N}}$ and a r.v. X.

- **Almost surely convergence:** $X_n \overset{\text{a.s.}}{\to} X$,
  - $\Leftrightarrow$ $\mathbb{P}(\lim_{n\to\infty} X_n = X) = 1$, meaning that for almost all ω, $X_n(\omega) \to X(\omega)$ as $n \to \infty$;
- **Converge in probability:** $X_n \overset{\text{p}}{\to} X$,
  - $\Leftrightarrow$ for all $\epsilon > 0$, $\lim_{n\to\infty} \mathbb{P}(|X_n - X| \geq \epsilon) = 0$;
- **Converge in L^p:** $X_n \overset{L^p}{\to} X$,
  - $\Leftrightarrow$ $\lim_{n\to\infty} \mathbb{E}\left[|X_n - X|^p\right] = 1$;
- **Converge in distribution:** $X_n \overset{d}{\to} X$,
  - $\Leftrightarrow$ $\lim_{n\to\infty} F_{X_n}(x) = F_X(x)$ for all x where $F(x)$ is continuous;

Examples and more properties are left as Bonus Homework.

随机变量收敛的主要类型

考虑一个随机变量序列 $\{X_n\}_{n\in\mathbb{N}}$ 和一个随机变量 X。

几乎处处收敛： $X_n \overset{\text{a.s.}}{\to} X$,

$\Leftrightarrow$ $\mathbb{P}(\lim_{n\to\infty} X_n = X) = 1$, meaning that for almost all ω, $X_n(\omega) \to X(\omega)$ as $n \to \infty$;

依概率收敛： $X_n \overset{\text{p}}{\to} X$,

$\Leftrightarrow$ for all $\epsilon > 0$, $\lim_{n\to\infty} \mathbb{P}(|X_n - X| \geq \epsilon) = 0$;

在 L^p：$X_n \overset{L^p}{\to} X$收敛,

$\Leftrightarrow$ $\lim_{n\to\infty} \mathbb{E}\left[|X_n - X|^p\right] = 1$;

依分布收敛： $X_n \overset{d}{\to} X$,

$\Leftrightarrow$ $\lim_{n\to\infty} F_{X_n}(x) = F_X(x)$对于所有满足 $F(x)$连续的 x；

更多示例和性质留作附加作业。

Weak LLN and strong LLN

Having established rigorous notions of convergence for random variables, we can now precisely formulate the Law of large numbers in two versions

Weak Law of large number (Khinchin's Law)

Given a collection of i.i.d. random variables $\{X_n\}$ with finite mean, the sample mean converges in probability to the expected value, that is for all $\epsilon > 0$

$$\lim_{n\to\infty} \mathbb{P}\left(\left|\frac{\sum_{i=1}^n X_i}{n} - \mathbb{E}[X_1]\right| \geq \epsilon\right) = 0.$$

Strong Law of large number (Kolmogorov's law)

Given a collection of i.i.d. random variables $\{X_n\}$ with finite mean, the sample mean converges almost surely to the expected value, that is

$$\mathbb{P}\left(\lim_{n\to\infty} \frac{\sum_{i=1}^n X_i}{n} = \mathbb{E}[X_1]\right) = 1.$$

弱大数定律和强大数定律

在为随机变量建立了严谨的收敛概念后，我们现在可以精确地表述大数定律的两个版本。

大数定律的弱形式（辛钦大数定律）

给定一组独立同分布且具有有限均值的随机变量 $\{X_n\}$，样本均值依概率收敛于期望值，即对于所有 $\epsilon > 0$

$$\lim_{n\to\infty} \mathbb{P}\left(\left|\frac{\sum_{i=1}^n X_i}{n} - \mathbb{E}[X_1]\right| \geq \epsilon\right) = 0.$$

大数定律的强形式（柯尔莫哥洛夫大数定律）

给定一组独立同分布且具有有限均值的随机变量 $\{X_n\}$，样本均值几乎必然收敛于期望值，即

$$\mathbb{P}\left(\lim_{n\to\infty} \frac{\sum_{i=1}^n X_i}{n} = \mathbb{E}[X_1]\right) = 1.$$

Proof of Weak LLN

We present a quick proof of the Weak LLN under the *additional assumption of finite variance*, that is $\mathbb{E}[X_1] = \mu$ and $\mathrm{Var}(X_1) = \sigma^2 < \infty$

- Introducing *Chebyshev's inequality*: Let X be a random variable with finite mean μ and finite non-zero variance σ^2. Then for any real number $k > 0$,

$$\mathbb{P}\Big(|X-\mu| \geq k\sigma\Big) \leq \frac{1}{k^2}.$$

- *Proof. It can be shown by*

$$\begin{aligned}\mathbb{P}\Big(|X-\mu| \geq k\sigma\Big) &= \mathbb{E}\Big[\mathbb{1}_{\{|X-\mu|\geq k\sigma\}}\Big] \\ &\leq \frac{1}{k^2\sigma^2}\mathbb{E}\Big[(X-\mu)^2\mathbb{1}_{\{|X-\mu|\geq k\sigma\}}\Big] \leq \frac{1}{k^2\sigma^2}\mathbb{E}\big[(X-\mu)^2\big] \\ &= \frac{\sigma^2}{k^2\sigma^2} = \frac{1}{k^2}.\end{aligned}$$

弱大数定律的证明

我们给出在有限方差的附加假设下Weak LLN的快速证明，即 $\mathbb{E}[X_1] = \mu$ 且Var(X_1) $= \sigma^2 < \infty$

介绍切比雪夫不等式：设 X 是一个具有有限均值 μ 和有限非零方差 σ^2的随机变量。那么对于任意实数 $k > 0$,

$$\mathbb{P}\Big(|X-\mu| \geq k\sigma\Big) \leq \frac{1}{k^2}.$$

证明。可以证明

$$\begin{aligned}\mathbb{P}\Big(|X-\mu| \geq k\sigma\Big) &= \mathbb{E}\Big[\mathbb{1}_{\{|X-\mu|\geq k\sigma\}}\Big] \\ &\leq \frac{1}{k^2\sigma^2}\mathbb{E}\Big[(X-\mu)^2\mathbb{1}_{\{|X-\mu|\geq k\sigma\}}\Big] \leq \frac{1}{k^2\sigma^2}\mathbb{E}\big[(X-\mu)^2\big] \\ &= \frac{\sigma^2}{k^2\sigma^2} = \frac{1}{k^2}.\end{aligned}$$

Proof of Weak LLN continued

- Denote $\overline{X}_n$ as the sample mean, and we have

$$\mathbb{E}[\overline{X}_n] = \mu, \text{ and } \mathrm{Var}(\overline{X}_n) = \frac{\sigma^2}{n};$$

- Using Chebyshev's inequality on $\overline{X}_n$, we have for all $\epsilon > 0$,

$$\mathbb{P}\left(|\overline{X}_n - \mu| \geq \varepsilon\right) \leq \frac{\sigma^2}{n\varepsilon^2},$$

and therefore,

$$\lim_{n\to\infty} \mathbb{P}\left(|\overline{X}_n - \mu| \geq \varepsilon\right) = 0,$$

which proves the weak LLN.

弱大数定律证明继续

记 X_n 为样本均值，我们有

$$\mathbb{E}[\overline{X}_n] = \mu, \text{ and } \mathrm{Var}(\overline{X}_n) = \frac{\sigma^2}{n};$$

在 X_n 上使用切比雪夫不等式，对于所有 $\epsilon > 0$,

$$\mathbb{P}\left(|\overline{X}_n - \mu| \geq \varepsilon\right) \leq \frac{\sigma^2}{n\varepsilon^2},$$

因此，

$$\lim_{n\to\infty} \mathbb{P}\left(|\overline{X}_n - \mu| \geq \varepsilon\right) = 0,$$

这证明了弱大数定律。

Outline

大纲

Outline

1. Bernoulli process
   - Definition: counting success
   - Distributions associated with Bernoulli process
   - Law of large number
   - **Convergence of random variables*
2. Poisson process
   - Continuous arrivals and definition
   - Distributions associated with Poisson process
   - Superposition and thinning
3. Simple random walk and Brownian motion
   - Simple random walk and continuous limit
   - Brownian motion
   - Central limit theorem

1. 伯努利过程
   - 定义：计数成功
   - 与伯努利过程相关的分布
   - 大数定律
   - *随机变量收敛
2. 泊松过程
   - 连续到达和定义
   - 与泊松过程相关的分布
   - 叠加和稀疏
3. 简单随机游走和布朗运动
   - 简单随机游走和连续极限
   - 布朗运动
   - 中心极限定理

Counting process

Based on the Bernoulli process $\{X_n\}$, we can define a corresponding counting process $\{S_n\}$ that counts the number of successes in a specified number of trials. This process is given by

$$S_k = \sum_{i=1}^{k} X_i.$$

- Counting process is a process $\{N_t\}$ such that
  - $N_0 = 0$;
  - N_t is increasing in t;
  - N_t increases by one every time that it changes.
- The counting process generated by a Bernoulli process can only be observed at integer time points $t = 0, 1, 2, \cdots$. How can we extend this to include more general time intervals, such as $t = 0.5$?

基于伯努利过程 $\{X_n\}$，我们可以定义一个相应的计数过程 $\{S_n\}$，该过程统计在指定次数试验中的成功次数。该过程表示为

$$S_k = \sum_{i=1}^{k} X_i.$$

计数过程是一个过程 $\{N_t\}$，使得

$N_0 = 0$;
N_t 在 t 中是递增的;
N_t 每次变化时增加一个。

由伯努利过程产生的计数过程只能观察到在整数时间点 $t = 0\ 1\ 2\ \cdots$, , , 。我们如何将其扩展以包含更一般的时间间隔，例如 $t = 0.5$?

Generalization of the counting process generated by a Bernoulli process

We would like the generalized path $\{S_t^{(m)}\}$ such that:

- it should be observable at smaller intervals; for example, we want to properly define $S^{(m)}_{\frac{k}{m}}$;
- it should maintain the same expected value at corresponding time points to prevent any divergence, that is $\mathbb{E}[S_n^{(m)}] = \mathbb{E}[S_n]$;

A natural generalization can be defined as follow:

- first, consider a Bernoulli process denoted as $\{X_n^{(m)}\}$ where each trial $X_n^{(m)}$ follows Bern$(\frac{p}{m})$;
- next, define the counting process $\{S^{(m)}_{\frac{n}{m}}\}$ by

$$S^{(m)}_{\frac{k}{m}} = \sum_{i=1}^{k} X_i^{(m)}, \text{ that is } S^{(m)}_{\frac{k}{m}} \sim \mathrm{B}\left(k, \frac{p}{m}\right)$$

- the conditions specified in the previous two points can be verified.

我们希望得到广义路径 $\{S_t^{(m)}\}$，使得:

它应该在小的时间间隔内可观测；例如，我们想要正确定义 $S^{(m)}_{\frac{k}{m}}$;

它应该在相应的时间点上保持相同的期望值以防止任何发散，即 $\mathbb{E}[S_n^{(m)}] = \mathbb{E}[S_n]$;

一种自然的推广可以定义如下:

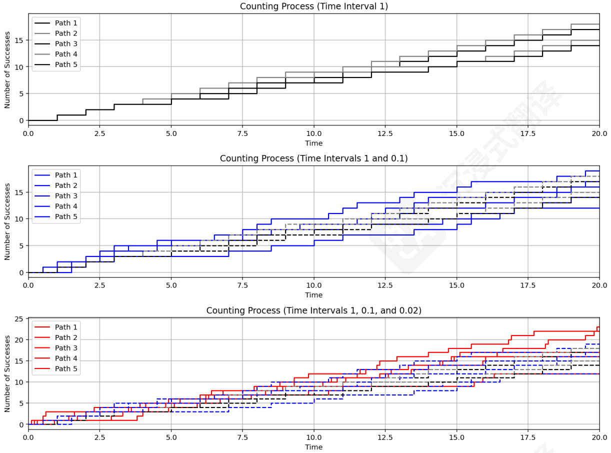
首先，考虑一个伯努利过程，记作 $\{X_n^{(m)}\}$，其中每个试验 $X_n^{(m)}$服从伯努利分布$\frac{p}{m}$;

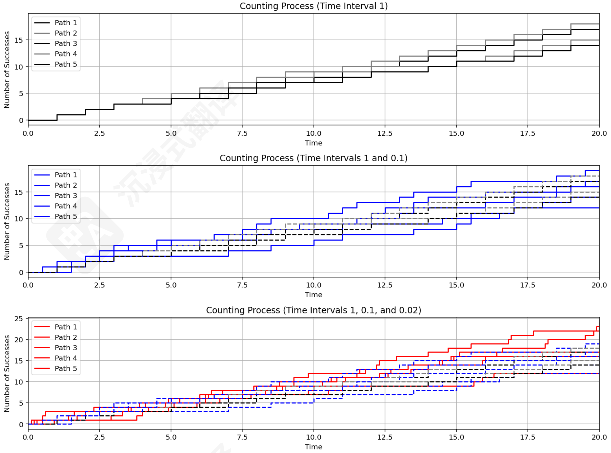
其次，定义计数过程 $\{S^{(m)}_{\frac{n}{m}}\}$ by

$$S^{(m)}_{\frac{k}{m}} = \sum_{i=1}^{k} X_i^{(m)}, \text{ that is } S^{(m)}_{\frac{k}{m}} \sim \mathrm{B}\left(k, \frac{p}{m}\right)$$

前两点中指定的条件可以验证。

Illustration of $\{S_t^{(m)}\}$





$m \to \infty$: Poisson process

To generalize the process to continuous time arrivals,, we take the limit $m \to \infty$ and obtain a limiting process $\{S_t^{\text{lim}}\}$.

- *Since $\{S_t^{(m)}\}$ is only defined on the rationals, extending the limit to irrational values of t requires care. We need to assume that the limiting process $\{S_t^{lim}\}$ is right-continuous with left limits (càdlàg)."*
- Formally, let us first check the limiting distribution of $S_1^{(m)}$, that is the limiting distribution of $\mathrm{B}(m, \frac{p}{m})$:

$$\begin{aligned}
\mathbb{P}(S_1^{\text{lim}} = k) &= \lim_{m\to\infty} \mathbb{P}(S_1^{(m)} = k) = \lim_{m\to\infty} C_m^k \left(\frac{p}{m}\right)^k \left(1 - \frac{p}{m}\right)^{m-k} \\
&= \lim_{m\to\infty} \frac{m!}{(m-k)!k!} \cdot \frac{p^k}{m^k} \left(1 - \frac{p}{m}\right)^m \cdot \left(1 - \frac{p}{m}\right)^{-k} \\
&= \frac{p^k}{k!} \lim_{m\to\infty} \left(1 - \frac{p}{m}\right)^m \cdot \frac{m(m-1)\cdots(m-k+1)}{(m-p)^k} \\
&= \frac{p^k}{k!} e^{-p}.
\end{aligned}$$

为了将过程推广到连续时间的到达，我们取极限$m \to \infty$，得到一个极限过程 $\{S_t^{\text{lim}}\}$。

由于 $\{S_t^{(m)}\}$ 仅在有理数上定义，将极限扩展到 t 的无理数值需要小心。我们需要假设极限过程 $\{S_t^{lim}\}$ 是右连续的，具有左极限（c`adl`ag）。”

形式上，让我们首先检查 $S_1^{(m)}$ 的极限分布，即B$(m, \frac{p}{m})$的极限分布：

$$\begin{aligned}
\mathbb{P}(S_1^{\text{lim}} = k) &= \lim_{m\to\infty} \mathbb{P}(S_1^{(m)} = k) = \lim_{m\to\infty} C_m^k \left(\frac{p}{m}\right)^k \left(1 - \frac{p}{m}\right)^{m-k} \\
&= \lim_{m\to\infty} \frac{m!}{(m-k)!k!} \cdot \frac{p^k}{m^k} \left(1 - \frac{p}{m}\right)^m \cdot \left(1 - \frac{p}{m}\right)^{-k} \\
&= \frac{p^k}{k!} \lim_{m\to\infty} \left(1 - \frac{p}{m}\right)^m \cdot \frac{m(m-1)\cdots(m-k+1)}{(m-p)^k} \\
&= \frac{p^k}{k!} e^{-p}.
\end{aligned}$$

Poisson distribution

This limiting distribution is so important that it is named after its inventor, the renowned French mathematician Siméon Poisson.

- X follows **Poisson distribution** with parameter λ, i.e. $X \sim \text{Pois}(\lambda)$ if
$$\mathbb{P}(X = k) = \frac{\lambda^k}{k!}e^{-k}, \text{ for } k \geq 0;$$
- $\mathbb{E}[X] = \lambda$, $\text{Var}(X) = \lambda$ and $M_X(t) = e^{\lambda(e^t - 1)}$.,
- The constraint $\lambda < 1$ used earlier for S_1^{lim} is not essential. More generally, by similar reasoning, one can show that $S_n^{\text{lim}} \sim \text{Pois}(p \cdot n)$ (the proof is left as homework);

这个极限分布非常重要，因此以发明者——著名的法国数学家 Sim´
eon Poisson 的名字命名。

X 服从参数为 λ的泊松分布，即 $X \sim \text{Pois}(\lambda)$如果

$$\mathbb{P}(X = k) = \frac{\lambda^k}{k!}e^{-k}, \text{ for } k \geq 0;$$

$\mathbb{E}[X] = \lambda$, $\text{Var}(X) = \lambda$ 并且 $M_X(t) = e^{\lambda(e^t - 1)}$。，

先前用于 S_1^{lim} 的约束 $\lambda < 1$ 并非必要。更一般地，通过类似的推理，可以证明 $S_n^{\text{lim}} \sim \text{Pois}(p \cdot n)$（证明留作作业）；

Characterizing the limiting process

Let us collect some useful properties of the limiting process $\{S_t^{\text{lim}}\}$ inherited from $\{S_t^{(m)}\}$:

- *independent increment*: In the discrete model, the increments from s to t, $S_t^{(m)} - S_s^{(m)}$, have a very natural property: the number of arrivals between times s and t depends only on the Bernoulli trials between times s and t, and is therefore independent of the arrivals before time s, that is

$$S_t^{(m)} - S_s^{(m)} \perp \{S_r^{(m)}\}_{r \leq s};$$

- *stationary increment*: In a Bernoulli process, the number of successes in trial numbers $n+1, \cdots, m$ and the number of successes in trial numbers $1, \cdots, m-n$ are both sums of $m-n$ independent Bernoulli variables. Therefore, these two numbers of successes have the same distribution, that is

$$S_t^{(m)} - S_s^{(m)} \stackrel{d}{=} S_{t-s}^{(m)}.$$

让我们收集一些从 $\{S_t^{(m)}\}$ 继承而来的极限过程 $\{S_t^{\text{lim}}\}$ 的有用性质：

独立增量：在离散模型中，从s 到 t、 $S_t^{(m)} - S_s^{(m)}$的增量具有一个非常自然的性质：在时间 s 和 t之间的到达次数仅取决于伯努利试验

在时间 s 和 t之间，并且因此独立于时间 s之前的到达，即

$$S_t^{(m)} - S_s^{(m)} \perp \{S_r^{(m)}\}_{r \leq s};$$

平稳增量：在一个伯努利过程中，试验编号 $n+1$、$\cdots$、m 中的成功次数与试验编号 1、$\cdots$、$m-n$ 中的成功次数都是 $m-n$ 个独立伯努利变量的和。因此，这两个成功次数具有相同的分布，即

$$S_t^{(m)} - S_s^{(m)} \stackrel{d}{=} S_{t-s}^{(m)}.$$

Poisson process

The two properties described above, combined with the distribution characterizations, completely determine the counting processes such that events arrive independently at a constant rate over time. We call counting processes of this kind **Poisson processes**, usually denoted by $\{N_t\}_{t\geq 0}$.

- A counting process $\{N_t\}_{t\geq 0}$ is called **a Poisson process with rate** $\lambda > 0$ if it satisfies the following properties: for all $s \leq t$,
  - $N_0 = 0$.
  - independent increment: $N_t - N_s \perp\!\!\!\perp \{N_r\}_{r\leq s}$;
  - stationary increments: $N_t - N_s \sim \text{Pois}(\lambda(t-s))$.
- The rate λ of a Poisson process can be interpreted precisely as the average number of arrivals per unit time since $\mathbb{E}[N_1] = \lambda$.

上述两种性质，再加上分布
特征，完全确定计数过程，使得事件以恒定速率独立地随时间到达。我们称

这种计数过程为泊松过程，通常表示为
$\{N_t\}_{t\geq 0}$。

一个计数过程 $\{N_t\}_{t\geq 0}$ 被称为具有速率的泊松过程
$\lambda > 0$ 如果它满足以下性质：对于所有 $s \leq t$,

$N_0 = 0$.
独立增量： $N_t - N_s \perp\!\!\!\perp \{N_r\}_{r\leq s}$；平稳增量：
$N_t - N_s \sim \text{Pois}(\lambda(t-s))$。

泊松过程的速率 λ 可以被精确地解释为单位时间内平均到达次数，
自 $\mathbb{E}[N_1] = \lambda$ 以来

Outline

1. Bernoulli process
   - Definition: counting success
   - Distributions associated with Bernoulli process
   - Law of large number
   - **Convergence of random variables*
2. Poisson process
   - Continuous arrivals and definition
   - Distributions associated with Poisson process
   - Superposition and thinning
3. Simple random walk and Brownian motion
   - Simple random walk and continuous limit
   - Brownian motion
   - Central limit theorem

大纲

1. 伯努利过程
   - 定义：计数成功
   - 与伯努利过程相关的分布
   - 大数定律
   - *随机变量收敛
2. 泊松过程
   - 连续到达和定义
   - 与泊松过程相关的分布
   - 叠加和稀疏
3. 简单随机游走和布朗运动
   - 简单随机游走和连续极限
   - 布朗运动
   - 中心极限定理

Exponential distribution: first arrival time

The **Exponential distribution** represents the first arrival time in a Poisson process, that is $T_1 \sim \text{Exp}(p)$:

- cdf: it can be derived from the structure of the Poisson process

$$F_{T_1}(t) := \mathbb{P}(T_1 \leq t) = \mathbb{P}(N_t > 0) = 1 - e^{-\lambda t}, \text{ for } t \geq 0,$$

  pdf: taking derivative of cdf, we get

$$f_{T_1}(t) = \lambda e^{-\lambda t}, \text{ for } t \geq 0;$$

- $\mathbb{E}[T_1] = \frac{1}{\lambda}$, $\text{Var}(T_1) = \frac{1}{\lambda^2}$ and $M_X(t) = \frac{\lambda}{\lambda - t}$,
- Memoryless property:

$$\mathbb{P}(T_1 > t | T_1 > s) = \mathbb{P}(T_1 > t - s),$$

  describes situations where previous failures or elapsed time do not affect future trials.
  - Also holds for $\text{Geo}(p)$.

指数分布表示泊松过程中的首次到达时间，即 $T_1 \sim \text{Exp}(p)$:

累积分布函数：它可以从泊松过程的结构中推导出来

$$F_{T_1}(t) := \mathbb{P}(T_1 \leq t) = \mathbb{P}(N_t > 0) = 1 - e^{-\lambda t}, \text{ for } t \geq 0,$$

概率密度函数：对累积分布函数求导，我们得到

$$f_{T_1}(t) = \lambda e^{-\lambda t}, \text{ for } t \geq 0;$$

$\mathbb{E}[T_1] = \frac{1}{\lambda}$, $\text{Var}(T_1) = \frac{1}{\lambda^2}$ and $M_X(t) = \frac{\lambda}{\lambda - t}$,
无记忆性：

$$\mathbb{P}(T_1 > t | T_1 > s) = \mathbb{P}(T_1 > t - s),$$

描述了在先前失败或经过的时间不影响未来试验的情况。

这也适用于 $\text{Geo}(p)$。

Gamma distribution: r-th arrival time

The **Gamma distribution** (also known as Erlang distribution) represents the r-th arrival time of , that is $T_r \sim \text{Gamma}(r, \lambda)$:

- cdf: it can be derived from the structure of the Poisson process

$$F_{T_r}(t) := \mathbb{P}(T_r \le t) = \mathbb{P}(N_t \ge r) = 1 - \sum_{i=0}^{r-1} \frac{(\lambda t)^i}{i!} e^{-\lambda t}, \text{ for } t \ge 0,$$

  pdf: taking derivative of cdf, we get

$$f_{T_r}(t) = \frac{\lambda^r t^{r-1} e^{-\lambda t}}{(r-1)!}, \text{ for } t \ge 0;$$

- $\mathbb{E}[T_r] = \frac{r}{\lambda}$, $\text{Var}(T_r) = \frac{r}{\lambda^2}$ and $M_{T_r}(t) = \left(\frac{\lambda}{\lambda - t}\right)^r$,
  - By analogy with the Negative Binomial distribution, these quantities can be calculated by observing that: *waiting times between arrivals are i.i.d.*

伽马分布（也称为厄兰分布）表示第 r 次到达时间，即 $T_r \sim \text{Gamma}(r, \lambda)$:

累积分布函数：它可以由泊松过程的结构推导出来

$$F_{T_r}(t) := \mathbb{P}(T_r \le t) = \mathbb{P}(N_t \ge r) = 1 - \sum_{i=0}^{r-1} \frac{(\lambda t)^i}{i!} e^{-\lambda t}, \text{ for } t \ge 0,$$

概率密度函数：对累积分布函数求导，我们得到

$$f_{T_r}(t) = \frac{\lambda^r t^{r-1} e^{-\lambda t}}{(r-1)!}, \text{ for } t \ge 0;$$

$\mathbb{E}[T_r] = \frac{r}{\lambda}$, $\text{Var}(T_r) = \frac{r}{\lambda^2}$ and $M_{T_r}(t) = \left(\frac{\lambda}{\lambda - t}\right)^r$,

与负二项分布类比，这些量可以通过观察到：到达之间的等待时间是独立同分布的来计算。

i.i.d. waiting times between arrivals

Let us consider the distribution of $T_1, T_2 - T_1$:

- for $t_1, t_2 > 0$, let us consider

$$\mathbb{P}(T_1 < t_1, T_2 < t_2) = \mathbb{P}\big(N(t_1) > 0, N(t_2) > 1\big)$$
$$= \begin{cases} \mathbb{P}(N(t_2) > 1), & \text{if } t_1 \geq t_2, \\ \mathbb{P}(N(t_1) = 1, N(t_2) > 1) + \mathbb{P}(N(t_1) > 1) & \text{if } t_1 < t_2. \end{cases}$$

Note that

$$\mathbb{P}(N(t_1) = 1, N(t_2) > 1) + \mathbb{P}(N(t_1) > 1)$$
$$= \mathbb{P}(N(t_1) = 1, N(t_2) - N(t_1) > 0) + \mathbb{P}(N(t_1) > 1)$$
$$= \lambda t_1 e^{-\lambda t_1}\big(1 - e^{-\lambda(t_2 - t_1)}\big) + 1 - e^{-\lambda t_1} - \lambda t_1 e^{-\lambda t_1}$$
$$= 1 - e^{-\lambda t_1} - \lambda t_1 e^{-\lambda t_2} \qquad \text{for } t_1 < t_2.$$

Then, the cdf of T_1, T_2 is given by

$$F_{T_1,T_2}(t_1, t_2) = \begin{cases} 1 - e^{-\lambda t_1} - \lambda t_1 e^{-\lambda t_1}, & \text{if } t_1 \geq t_2, \\ 1 - e^{-\lambda t_1} - \lambda t_1 e^{-\lambda t_2}, & \text{if } t_1 < t_2, \end{cases}$$

which leads to the pdf function

$$f_{T_1,T_2}(t_1, t_2) = \lambda^2 e^{-\lambda t_2} \qquad \text{for } t_1 < t_2.$$

让我们考虑 T_1 $T_2 - T_1$的分布：

对于 t_1 $t_2 > 0$, ，让我们考虑

$$\mathbb{P}(T_1 < t_1, T_2 < t_2) = \mathbb{P}\big(N(t_1) > 0, N(t_2) > 1\big)$$
$$= \begin{cases} \mathbb{P}(N(t_2) > 1), & \text{if } t_1 \geq t_2, \\ \mathbb{P}(N(t_1) = 1, N(t_2) > 1) + \mathbb{P}(N(t_1) > 1) & \text{if } t_1 < t_2. \end{cases}$$

请注意

$$\mathbb{P}(N(t_1) = 1, N(t_2) > 1) + \mathbb{P}(N(t_1) > 1)$$
$$= \mathbb{P}(N(t_1) = 1, N(t_2) - N(t_1) > 0) + \mathbb{P}(N(t_1) > 1)$$
$$= \lambda t_1 e^{-\lambda t_1}\big(1 - e^{-\lambda(t_2 - t_1)}\big) + 1 - e^{-\lambda t_1} - \lambda t_1 e^{-\lambda t_1}$$
$$= 1 - e^{-\lambda t_1} - \lambda t_1 e^{-\lambda t_2} \qquad \text{for } t_1 < t_2.$$

那么， T_1, T_2 的累积分布函数为

$$F_{T_1,T_2}(t_1, t_2) = \begin{cases} 1 - e^{-\lambda t_1} - \lambda t_1 e^{-\lambda t_1}, & \text{if } t_1 \geq t_2, \\ 1 - e^{-\lambda t_1} - \lambda t_1 e^{-\lambda t_2}, & \text{if } t_1 < t_2, \end{cases}$$

这导致概率密度函数

$$f_{T_1,T_2}(t_1, t_2) = \lambda^2 e^{-\lambda t_2} \qquad \text{for } t_1 < t_2.$$

i.i.d. waiting times between arrivals continued

The distribution of $T_1, T_2 - T_1$ continues:

- Using the formula of transformation of random variables (details at Slides 61), we have

$$f_{T_1,T_2-T_1}(t_1,t_2) = f_{T_1,T_2}(t_1,t_1+t_2) = \lambda^2 e^{-\lambda(t_1+t_2)}, \quad \text{for } t_1,t_2 > 0.$$

- Then calculating the marginal distribution, it is not hard to see that

$$f_{T_1}(t_1) = \lambda e^{-\lambda t_1} \text{ and } f_{T_2}(t_2) = \lambda e^{-\lambda t_2}, \text{ for } t_1,t_2 > 0,$$

which leads to the independence.

- We leave the case of $T_1, T_2 - T_1, T_3 - T_2$ as homework and in general, we can prove that $T_1, T_2 - T_1, \cdots, T_n - T_{n-1}$ are i.i.d.

变量 T_1 $T_2 - T_1$ 的分布继续：

使用随机变量变换公式（详情见幻灯片61），我们有

$$f_{T_1,T_2-T_1}(t_1,t_2) = f_{T_1,T_2}(t_1,t_1+t_2) = \lambda^2 e^{-\lambda(t_1+t_2)}, \quad \text{for } t_1,t_2 > 0.$$

然后计算边缘分布，不难看出

$$f_{T_1}(t_1) = \lambda e^{-\lambda t_1} \text{ and } f_{T_2}(t_2) = \lambda e^{-\lambda t_2}, \text{ for } t_1,t_2 > 0,$$

这导致了独立性。

我们将 T_1 $T_2 - T_1$ $T_3 - T_2$ 的情况留作作业，并且一般来说，我们可以证明 T_1 $T_2 - T_1$ $\cdots$ $T_n - T_{n-1}$，，是独立同分布的。

Allow r to take non-integer values

- Revisit the pdf of Gamma(r, λ) for integer r:

$$f_{T_r}(t) = \frac{\lambda^r t^{r-1} e^{-\lambda t}}{(r-1)!} = \lambda \cdot \frac{(\lambda t)^{r-1} e^{-\lambda t}}{(r-1)!},$$

  what if we allow r to take non-integer values?

- Since x^r is well-defined for non-integer values for positive x, we must extend the definition of $(r-1)!$ for non-integer r.
- Approach: Use the normalization condition $\int_{t \geq 0} f_{T_r}(t) \mathrm{d}t = 1$.
- Introducing Gamma functions $\Gamma(r)$ such that

$$\int_{t \geq 0} \lambda \cdot \frac{(\lambda t)^{r-1} e^{-\lambda t}}{\Gamma(r)} \mathrm{d}t = 1,$$

  that is

$$\Gamma(r) := \int_{t \geq 0} t^{r-1} e^{-t} \mathrm{d}t.$$

重新审视Gamma(r, λ)在整数 r情况下的概率密度函数:

$$f_{T_r}(t) = \frac{\lambda^r t^{r-1} e^{-\lambda t}}{(r-1)!} = \lambda \cdot \frac{(\lambda t)^{r-1} e^{-\lambda t}}{(r-1)!},$$

如果我们允许 r 取非整数值会怎样?

由于 x^r 对于正数 x在非整数值时已定义，我们必须扩展 $(r-1)!$ 的定义以适用于非整数 r。

方法：使用归一化条件 $\int_{t \geq 0} f_{T_r}(t) \mathrm{d}t = 1$。

介绍伽马函数 $\Gamma(r)$，使得

$$\int_{t \geq 0} \lambda \cdot \frac{(\lambda t)^{r-1} e^{-\lambda t}}{\Gamma(r)} \mathrm{d}t = 1,$$

那是

$$\Gamma(r) := \int_{t \geq 0} t^{r-1} e^{-t} \mathrm{d}t.$$

Allow r to take non-integer values

Properties of Gamma functions:

- For positive integers: $\Gamma(r) = (r-1)!$.
- Recurrence relation: $\Gamma(r+1) = r \cdot \gamma(r)$.
- Special values: $\Gamma(1) = 1, \Gamma(\frac{1}{2}) = \sqrt{\pi}$. (Leave as homework)

Generalize the definition of Gamma distribution $X \sim \text{Gamma}(r, \lambda)$:

- pdf: $f_X(x) = \frac{\lambda^r x^{r-1} e^{-\lambda x}}{\Gamma(r)}$;
- (cdf: no explicit formulation for general r;)
- $\mathbb{E}[T_r] = \frac{r}{\lambda}$, $\text{Var}(T_r) = \frac{r}{\lambda^2}$ and $M_{T_r}(t) = \left(\frac{\lambda}{\lambda - t}\right)^r$, remain the same.

伽马函数的性质：

对于正整数： $\Gamma(r) = (r-1)!$。

递推关系： $\Gamma(r+1) = r \cdot \gamma(r)$.

特殊值： $\Gamma(1) = 1$ $\Gamma(\frac{1}{2}) = \sqrt{\pi}$.，(留作作业)

推广Gamma分布 $X \sim \text{Gamma}(r, \lambda)$的定义：

pdf: $f_X(x) = \frac{\lambda^r x^{r-1} e^{-\lambda x}}{\Gamma(r)}$;

(cdf: 对于一般的 r没有显式公式；)
$\mathbb{E}[T_r] = \frac{r}{\lambda}$， $\text{Var}(T_r) = \frac{r}{\lambda^2}$和 $M_{T_r}(t) = \left(\frac{\lambda}{\lambda - t}\right)^r$保持不变。

38 / 76

Outline

1. Bernoulli process
   - Definition: counting success
   - Distributions associated with Bernoulli process
   - Law of large number
   - **Convergence of random variables*
2. Poisson process
   - Continuous arrivals and definition
   - Distributions associated with Poisson process
   - Superposition and thinning
3. Simple random walk and Brownian motion
   - Simple random walk and continuous limit
   - Brownian motion
   - Central limit theorem

大纲

1. 伯努利过程
   - 定义：计数成功
   - 与伯努利过程相关的分布
   - 大数定律
   - *随机变量收敛
2. 泊松过程
   - 连续到达和定义
   - 与泊松过程相关的分布
   - 叠加和稀疏
3. 简单随机游走和布朗运动
   - 简单随机游走和连续极限
   - 布朗运动
   - 中心极限定理

Independent sum of Poisson random variable

We start by a simple property of Poisson random variables:

- Consider $X \sim \text{Pois}(\lambda)$ and $Y \sim \text{Pois}(\mu)$ are two independent random variables, what is the distribution of $X+Y$?
- Brutal calculation:

$$\mathbb{P}(X+Y=k)=\sum_{i=0}^{k}\mathbb{P}(X=i,Y=k-i)=\sum_{i=0}^{k}\mathbb{P}(X=i)\mathbb{P}(Y=k-i)$$
$$=\sum_{i=0}^{k}\frac{\lambda^i}{i!}e^{-\lambda}\frac{\mu^{k-i}}{(k-i)!}e^{-\mu}=\sum_{i=0}^{k}\frac{1}{i!(k-i)!}\lambda^i\mu^{k-i}e^{-\lambda-\mu}$$
$$=\frac{(\lambda+\mu)^k e^{-\lambda-\mu}}{k!}\sum_{i=0}^{k}\frac{k!}{i!(k-i)!}\left(\frac{\lambda}{\lambda+\mu}\right)^i\left(\frac{\mu}{\lambda+\mu}\right)^{k-i}=\frac{(\lambda+\mu)^k e^{-\lambda-\mu}}{k!},$$

therefore, $X+Y \sim \text{Pois}(\lambda+\mu)$.

- Another approach: For a Poisson process $\{N_t\}$ with rate 1, note that $(X,Y)\stackrel{d}{=}(N_\lambda, N_{\lambda+\mu}-N_\lambda)$ implies $X+Y\stackrel{d}{=}N_{\lambda+\mu}$, which gives the desired conclusion.

我们首先从一个泊松随机变量的简单性质开始:

考虑 $X \sim \text{Pois}(\lambda)$和 $Y \sim \text{Pois}(\mu)$是两个独立的随机变量,
$X+Y$?的分布是什么
暴力计算:

$$\mathbb{P}(X+Y=k)=\sum_{i=0}^{k}\mathbb{P}(X=i,Y=k-i)=\sum_{i=0}^{k}\mathbb{P}(X=i)\mathbb{P}(Y=k-i)$$
$$=\sum_{i=0}^{k}\frac{\lambda^i}{i!}e^{-\lambda}\frac{\mu^{k-i}}{(k-i)!}e^{-\mu}=\sum_{i=0}^{k}\frac{1}{i!(k-i)!}\lambda^i\mu^{k-i}e^{-\lambda-\mu}$$
$$=\frac{(\lambda+\mu)^k e^{-\lambda-\mu}}{k!}\sum_{i=0}^{k}\frac{k!}{i!(k-i)!}\left(\frac{\lambda}{\lambda+\mu}\right)^i\left(\frac{\mu}{\lambda+\mu}\right)^{k-i}=\frac{(\lambda+\mu)^k e^{-\lambda-\mu}}{k!},$$

因此, $X+Y \sim \text{Pois}(\lambda+\mu)$。

另一种方法:对于泊松过程 $\{N_t\}$ (速率 1), 注意到 $(X,Y)\stackrel{d}{=}(N_\lambda, N_{\lambda+\mu}-N_\lambda)$ 意味着 $X+Y\stackrel{d}{=}N_{\lambda+\mu}$, 这给出了所需的结论。

Independent sum of Poisson process: an illustrative example

To illustrate the independent sum of Poisson process, let us consider a simple example.

- Both men and women shop at a certain store (for example, a lingerie store), albeit at different rate. On average, μ men and λ women arrive per hour. The arrivals of men and women are modelled by independent Poisson processes. What can we say about the process that counts the total arrivals of all customers at the store?
- Define
$$M_t = \# \text{ men that arrived by time } t,$$
$$N_t = \# \text{ women that arrived by time } t,$$
so that $\{M_t\}_{t\geq 0}$ is a Poisson process with rate μ and $\{N_t\}_{t\geq 0}$ is a Poisson process with rate λ.
- *Independent process:* $\{M_t\}_{t\geq 0} \perp\!\!\!\perp \{N_t\}_{t\geq 0}$ means :
$$(M_{t_1}, \cdots, M_{t_k}) \perp\!\!\!\perp (N_{\tilde{t}_1}, \cdots, N_{\tilde{t}_\ell}), \text{ for all } \{t_i\} \text{ and } \{\tilde{t}_j\}.$$

为了说明泊松过程的独立和，让我们考虑一个简单的例子。

- 男性和女性都会去某个商店（例如，内衣店），尽管他们的到访速率不同。平均而言， μ 名男性每小时和 λ 名女性每小时到达。男性和女性的到访被建模为独立的泊松过程。我们能对计算商店所有顾客总到访数的进程说什么吗?
- 定义
$$M_t = \text{名在时间 } t \text{之前到达的男性},$$
$$N_t = \# \text{ 在时间 } t \text{ 到达的女性},$$
因此 $\{M_t\}_{t\geq 0}$ 是一个速率为 μ 的泊松过程，且 $\{N_t\}_{t\geq 0}$ 是一个速率为 λ 的泊松过程。
- 独立过程： $\{M_t\}_{t\geq 0} \perp\!\!\!\perp \{N_t\}_{t\geq 0}$ 表示：
$$(M_{t_1}, \cdots, M_{t_k}) \perp\!\!\!\perp (N_{\tilde{t}_1}, \cdots, N_{\tilde{t}_\ell}), \text{ for all } \{t_i\} \text{ and } \{\tilde{t}_j\}.$$

Illustrative example continued

- Define

$$R_t = \#\text{customers that arrived by time } t = M_t + N_t,$$

  the total number of customers.
- What can we say about the properties of the process $\{R_t\}_{t\geq 0}$?
  - Note that

$$R_t - R_s = (M_t - M_s) + (N_t - N_s).$$

    As $\{M_t\}_{t\geq 0}$ is a Poisson process, $M_t - M_s \perp\!\!\!\perp \{M_u\}_{u\leq s}$, while $M_t - M_s \perp\!\!\!\perp \{N_u\}_{u\leq s}$ as $\{M_t\}_{t\geq 0} \perp\!\!\!\perp \{N_t\}_{t\geq 0}$. Exactly the same reasoning applies to $N_t - N_s$. Therefore,

$$R_t - R_s \perp\!\!\!\perp \{R_u\}_{u\leq s}.$$

  - Similarly, note that $M_t - M_s$ and $N_t - N_s$ are independent Poisson random variables with means $\mu(t-s)$ and $\lambda(t-s)$. Therefore,

$$R_t - R_s \sim \text{Pois}\big((\mu+\lambda)(t-s)\big).$$

定义

$$R_t = \text{在时间 } t = M_t + N_t\text{前到达的顾客},$$

顾客总数。

关于过程 $\{R_t\}_{t\geq 0}$? 的性质，我们能说什么?

请注意

$$R_t - R_s = (M_t - M_s) + (N_t - N_s).$$

由于 $\{M_t\}_{t\geq 0}$ 是泊松过程，$M_t - M_s \perp\!\!\!\perp \{M_u\}_{u\leq s}$，而 $M_t - M_s \perp\!\!\!\perp \{N_u\}_{u\leq s}$ 作为 $\{M_t\}_{t\geq 0} \perp\!\!\!\perp \{N_t\}_{t\geq 0}$。完全相同的推理也适用于 $N_t - N_s$。因此,

$$R_t - R_s \perp\!\!\!\perp \{R_u\}_{u\leq s}.$$

同样，请注意 $M_t - M_s$ 和 $N_t - N_s$ 是独立的泊松随机变量，其均值分别为 $\mu(t-s)$ 和 $\lambda(t-s)$。因此,

$$R_t - R_s \sim \text{Pois}\big((\mu+\lambda)(t-s)\big).$$

Superposition principle

- Thus we have shown that $\{R_t\}_{t\geq 0}$ is a Poisson process with rate $\mu+\lambda$! This is intuitive: if men arrive at an average rate of μ per hour, and women arrive at a rate of λ per hour, then on average $\mu+\lambda$ customers arrive per hour.
- More generally, we have the following **principle of superposition**:

Principle of superposition

If $\{N^{(1)}\}_t, \cdots, \{N^{(k)}\}_t$ are independent Poisson processes with rates $\mu_1, \cdots, \mu_k$, then $\{N^{(1)}+\cdots+N^{(k)}\}_t$ is a Poisson process with rate $\mu_1+\cdots+\mu_k$.

因此我们证明了 $\{R_t\}_{t\geq 0}$ 是一个速率为$\mu+\lambda$! 的泊松过程。这是直观的：如果男性以每小时 μ 的平均速率到达，而女性以每小时 λ 的速率到达，那么平均每小时将有$\mu+\lambda$ 名顾客到达。

更一般地，我们有以下叠加原理：

叠加原理

如果 $\{N^{(1)}\}_t$ $\cdots$ $\{N^{(k)}\}_t$, , 是独立的泊松过程，其速率分别为 μ_1 $\cdots$ μ_k, , , 那么 $\{N^{(1)}+\cdots+N^{(k)}\}_t$是一个速率等于$\mu_1+\cdots+\mu_k$的泊松过程。

Thinning principle

The principle of thinning is the precise opposite of superposition. To illustrate it, let us reconsider the above example from a different perspective.

- On average, μ customers arrive at a store per hour. Each customer is independently) a man with probability p and a woman with probability $1-p$. What can we say about the processes that count the arrivals of men and of women at the store?
- Let us make some careful definitions.
  - Set
    $$R_t = \# \text{ customers that arrived by time } t,$$
    be a Poisson process with rate μ.
  - Set
    $$X_k = \text{ gender of customer } k,$$
    be i.i.d. with $\mathbb{P}(X_k = \text{“}m\text{”}) = p$ and $\mathbb{P}(X_k = \text{“}w\text{”}) = 1-p$.

稀疏原理与叠加原理正好相反。为了说明这一点，让我们从不同的角度重新考虑上述例子。

平均而言， μ名顾客每小时到达商店。每位顾客是独立地) 男性，概率为 p ，女性，概率为 $1-p$. 关于计数那些过程，我们能说什么?在商店中，男性和女性的到达次数如何?

让我们做一些仔细的定义。

Set
$$R_t = \# \text{ customers that arrived by time } t,$$
是一个泊松过程，速率为 μ。

Set
$$X_k = \text{ gender of customer } k,$$
独立同分布，具有 $\mathbb{P}(X_k = \text{“}m\text{”}) = p$ 和 $\mathbb{P}(X_k = \text{“}w\text{”}) = 1-p$。

Illustrative example

- With these definitions, we have

$$M_t = \#\text{ men that arrived by time } t = \sum_{k=1}^{R_t} \mathbb{1}\{X_k = \text{“}m\text{”}\},$$

$$N_t = \#\text{ women that arrived by time } t = \sum_{k=1}^{R_t} \mathbb{1}\{X_k = \text{“}w\text{”}\}.$$

- For sake of illustration, let us compute the distribution of M_t by law of total expectation,

$$\mathbb{P}(M_t = k) = \mathbb{E}\big[\mathbb{P}(M_t = k | R_t)\big] = \sum_{r=k}^{\infty} \mathbb{P}(M_t = k | R_t = r)\mathbb{P}(R_t = r).$$

- It can be shown that $M_t | R_t = r \ \sim \mathrm{B}(r, p)$, and therefore

$$\mathbb{P}(M_t = k) = \sum_{r=k}^{\infty} \binom{r}{k} p^k (1-p)^{r-k} \frac{(\mu t)^r}{r!} e^{-\mu t} = \frac{p^k (\mu t)^k e^{-\mu t}}{k!} \sum_{r=k}^{\infty} \frac{(\mu(1-p)t)^{r-k}}{(r-k)!}$$
$$= \frac{(\mu p t)^k e^{-\mu t}}{k!} e^{(1-p)\mu t} = \frac{(\mu p t)^k}{k!} e^{-p\mu t}.$$

有了这些定义，我们就有

$$M_t = \#\text{ men that arrived by time } t = \sum_{k=1}^{R_t} \mathbb{1}\{X_k = \text{“}m\text{”}\},$$

$$N_t = \#\text{ women that arrived by time } t = \sum_{k=1}^{R_t} \mathbb{1}\{X_k = \text{“}w\text{”}\}.$$

为了说明，让我们通过全期望法则计算 M_t 的分布，

$$\mathbb{P}(M_t = k) = \mathbb{E}\big[\mathbb{P}(M_t = k | R_t)\big] = \sum_{r=k}^{\infty} \mathbb{P}(M_t = k | R_t = r)\mathbb{P}(R_t = r).$$

可以证明 $M_t | R_t = r \ \sim \mathrm{B}(r, p)$，因此

$$\mathbb{P}(M_t = k) = \sum_{r=k}^{\infty} \binom{r}{k} p^k (1-p)^{r-k} \frac{(\mu t)^r}{r!} e^{-\mu t} = \frac{p^k (\mu t)^k e^{-\mu t}}{k!} \sum_{r=k}^{\infty} \frac{(\mu(1-p)t)^{r-k}}{(r-k)!}$$
$$= \frac{(\mu p t)^k e^{-\mu t}}{k!} e^{(1-p)\mu t} = \frac{(\mu p t)^k}{k!} e^{-p\mu t}.$$

Illustrative example continued

- To conclude, $M_t \sim$ Pois$(p\mu t)$.
- In fact, we can easily check that $\{M_t\}_{t\geq 0}$ has stationary independent increments, so that the counting process that counts only the men $\{M_t\}_{t\geq 0}$ is itself a Poisson process with rate $(p\mu)$!
- This is very intuitive: if customers arrive at an average rate of μ per hour, and each customer is a man with probability p, then there should be on average $p\mu$ men arriving per hour.
- In precisely the same way, we find that the number of women $\{N_t\}_{t\geq 0}$ is a Poisson process with rate $(1-p)\mu$.

总之， $M_t \sim$ 概率Pois$(p\mu t)$.

实际上，我们可以轻易地验证 $\{M_t\}_{t\geq 0}$ 是平稳的
在事实上，我们可以很容易地验证 $\{M_t\}_{t\geq 0}$ 具有平稳独立增量，因此仅计数男性 $\{M_t\}_{t\geq 0}$ 的计数过程本身就是一个泊松过程，速率为 $(p\mu)$!

这是非常直观的：如果顾客以每小时 μ 的平均速率到达，并且每个顾客是男性的概率为 p，那么每小时应该平均有 $p\mu$ 名男性到达。

恰恰如此，我们发现女性数量 $\{N_t\}_{t\geq 0}$ 是一个以速率 $(1-p)\mu$ 的泊松过程。

Illustrative example continued

- We can say even more: the processes $\{M_t\}_{t\geq 0}$ and $\{N_t\}_{t\geq 0}$ are in fact independent! Let us verify, for example, that M_t and N_t are independent:

$$\begin{aligned}
&\mathbb{P}(M_t = m, N_t = n) = \mathbb{P}(M_t = m, R_t = m+n) \\
&= \mathbb{P}(R_t = m+n) \cdot \mathbb{P}(M_t = m | R_t = m+n) \\
&= \frac{(\mu t)^{m+n}}{(m+n)!} e^{-\mu t} \cdot \binom{m+n}{n} p^m (1-p)^n = \frac{(p\mu t)^m}{m!} e^{-p\mu t} \cdot \frac{((1-p)\mu t)^n}{n!} e^{-(1-p)\mu t} \\
&= \mathbb{P}(M_t = m) \cdot \mathbb{P}(N_t = n) \text{ for all } m, n.
\end{aligned}$$

- More generally, we have the following **principle of thinning**:

Principle of thinning

If N_t is a Poisson process with rate μ, and if each arrival is independently assigned one of k labels with probabilities $p_1, \cdots, p_k$, then the counting processes $\{N_t^{(1)}\}, \cdots, \{N_t^{(k)}\}$ that count the arrivals of each type are independent Poisson processes with rates $p_1\mu, \cdots, p_k\mu$.

我们甚至可以说更多：过程 $\{M_t\}_{t\geq 0}$ 和 $\{N_t\}_{t\geq 0}$ 实际上是独立的！让我们验证一下，例如， M_t 和 N_t 是否独立：

$$\begin{aligned}
&\mathbb{P}(M_t = m, N_t = n) = \mathbb{P}(M_t = m, R_t = m+n) \\
&= \mathbb{P}(R_t = m+n) \cdot \mathbb{P}(M_t = m | R_t = m+n) \\
&= \frac{(\mu t)^{m+n}}{(m+n)!} e^{-\mu t} \cdot \binom{m+n}{n} p^m (1-p)^n = \frac{(p\mu t)^m}{m!} e^{-p\mu t} \cdot \frac{((1-p)\mu t)^n}{n!} e^{-(1-p)\mu t} \\
&= \mathbb{P}(M_t = m) \cdot \mathbb{P}(N_t = n) \text{ for all } m, n.
\end{aligned}$$

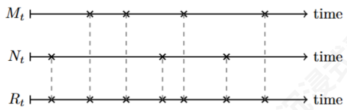
更一般地，我们有以下稀疏原则：

稀疏原则

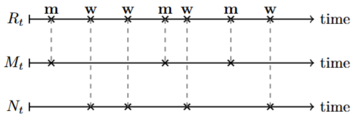
如果 N_t 是一个速率为 μ 的泊松过程，并且如果每个到达被独立地分配一个来自 k 标签的标签，其概率为 $p_1 \cdots p_k$、$p_1 \cdots p_k$、$p_1 \cdots p_k$，那么计数过程 $\{N_t^{(1)}\} \cdots \{N_t^{(k)}\}$、$\{N_t^{(1)}\} \cdots \{N_t^{(k)}\}$、$\{N_t^{(1)}\} \cdots \{N_t^{(k)}\}$，它们分别计算每种类型的到达，都是独立的泊松过程，其速率为 $p_1\mu \cdots p_k\mu$、$p_1\mu \cdots p_k\mu$、$p_1\mu \cdots p_k\mu$。

Superposition and thinning principle

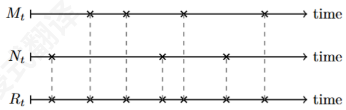
- Illustrating of principle of superposition:



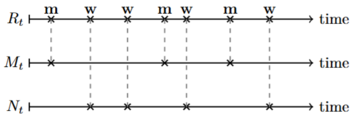
- Illustrating of principle of thinning:



叠加原理说明:



稀疏原理说明:



Outline

概要

Outline

1. Bernoulli process
   - Definition: counting success
   - Distributions associated with Bernoulli process
   - Law of large number
   - **Convergence of random variables*
2. Poisson process
   - Continuous arrivals and definition
   - Distributions associated with Poisson process
   - Superposition and thinning
3. Simple random walk and Brownian motion
   - Simple random walk and continuous limit
   - Brownian motion
   - Central limit theorem

概要

1. 伯努利过程
   - 定义：计数成功
   - 与伯努利过程相关的分布
   - 大数定律
   - **随机变量收敛*
2. 泊松过程
   - 连续到达和定义
   - 与泊松过程相关的分布
   - 叠加和稀疏
3. 简单随机游走和布朗运动
   - 简单随机游走和连续极限
   - 布朗运动
   - 中心极限定理

Simple random walk

A *random walk* is a stochastic process that describes a path that consists of a succession of random steps.

- Mathematical definition: A **random walk** $\{S_n\}_{n\geq 0}$ is the random process
$$S_n = S_0 + X_1 + ... + X_n,$$
where $X_1, X_2, \cdots$ are i.i.d. random variables independent of S_0.
- If S_0 is integer and the steps X_i only take the values ± 1, the random walk is called **simple**. **A simple random walk** is called **symmetric** if $\mathbb{P}(X_i = 1) = \mathbb{P}(X_i = -1) = \frac{1}{2}$.
- We will restrict our discussion on symmetric simple random walk.
  - Properties for symmetric simple random walk: $\mathbb{E}[S_n] = 0$ and $\mathrm{Var}(S_n) = n$ when $S_0 = 0$.

随机游走是一个随机过程，描述了一条由一系列随机步组成的路径。

数学定义：一个随机游走 $\{S_n\}_{n\geq 0}$ 是一个随机过程
$$S_n = S_0 + X_1 + ... + X_n,$$
其中 $X_1, X_2, \cdots$ 是独立同分布的随机变量，且独立于 S_0。

如果 S_0 是整数，并且步长 X_i 只取值 ± 1，那么随机游走被称为简单随机游走。如果 $\mathbb{P}(X_i = 1) = \mathbb{P}(X_i = -1) = \frac{1}{2}$，则简单随机游走被称为对称的。

我们将限制讨论范围于对称简单随机游走。

对称简单随机游走的性质：
$\mathbb{E}[S_n] = 0$ and $\mathrm{Var}(S_n) = n$ when $S_0 = 0$.

Some interesting questions

There are several interesting quantities and questions we would like to investigate later (at Chapter 3.)

- Hitting time $T_{a,b}$ of a barrier $[a, b]$ where $a < S_0 < b$:

$$T_{a,b} := \min\{n \geq 0 : S_n = a \text{ or } S_n = b\}.$$

- Question 1: Will we ever hit a boundary?
- Question 2: Which boundary will we hit?
- Question 3: How long will it take?
- We will answer all of these in Chapter 3 using first step analysis.

我们希望稍后（在第三章）研究几个有趣的量和问题。

击中时间 $T_{a,b}$ 的障碍物 $[a, b]$，其中 $a < S_0 < b$:

$$T_{a,b} := \min\{n \geq 0 : S_n = a \text{ or } S_n = b\}.$$

问题 1：我们是否永远会击中边界?

问题 2：我们会击中哪个边界?

问题 3：需要多长时间?

我们将使用第一步分析在第3章中回答所有这些问题。

Continuous time limit

Recall our discussion on continuous time generalization of the counting process:

- We would like the generalized path $\{S_t^{(m)}\}$ such that:
  - we want to properly define $S_{\frac{k}{m}}^{(m)}$;
  - it should maintain the same expected value, that is $\mathbb{E}[S_n^{(m)}] = \mathbb{E}[S_n]$;

Similarly, we would like our continuous time generalization of simple random walk to behave as:

- it should be observable at smaller intervals; for example, we want to properly define $S_{\frac{k}{N}}^{(N)}$;
- it should maintain the same expected value *and variance* at corresponding time points to prevent any divergence, that is

$$\mathbb{E}[S_n^{(N)}] = \mathbb{E}[S_n], \quad \mathrm{Var}(S_n^{(N)}) = \mathrm{Var}(S_n),$$

回顾我们对计数过程连续时间推广的讨论

过程:

我们希望广义路径 $\{S_t^{(m)}\}$ 满足:

我们想要正确定义 $S_k^{(m)}$;$_m$

它应该保持相同的预期值，即$\mathbb{E}[S_n^{(m)}] = \mathbb{E}[S_n]$;

同样地，我们也希望我们的简单连续时间推广
随机游走以表现:

它应该在小的时间间隔内可观察；例如，我们想要正确地定义
$S_k^{(N)}$;$_N$

它应该在相应的时间点保持相同的期望值和方差，以防止任何发散，即

$$\mathbb{E}[S_n^{(N)}] = \mathbb{E}[S_n], \quad \mathrm{Var}(S_n^{(N)}) = \mathrm{Var}(S_n),$$

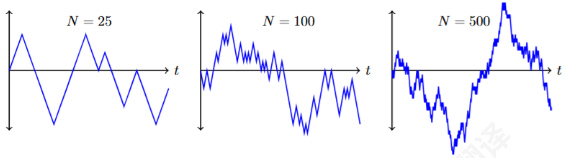
Continuous time limit

A natural generalization can be defined as follow:

- consider a symmetric simple random walk, denoted as $\{X_i^{(N)}\}$ where $\mathbb{P}(X_i^{(N)} = 1) = \mathbb{P}(X_i^{(N)} = -1) = \frac{1}{2}$ and $X_i^{(N)}$ are i.i.d.;
- next, define the limiting process $\{S_{\frac{n}{N}}^{(N)}\}$ by

$$S_{\frac{k}{N}}^{(N)} = \frac{1}{\sqrt{N}} \sum_{i=1}^{k} X_i^{(N)};$$

the scaling coefficient $\frac{1}{\sqrt{N}}$ is to maintain the same variance.



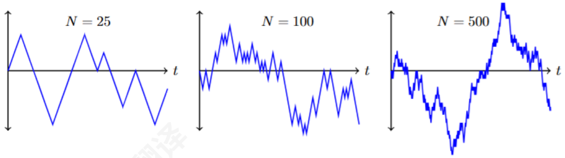
一个自然的推广可以定义如下:

考虑一个对称的简单随机游走，记作 $\{X_i^{(N)}\}$，其中 $\mathbb{P}(X_i^{(N)} = 1) = \mathbb{P}(X_i^{(N)} = -1) = \frac{1}{2}$ 和 $X_i^{(N)}$ 是独立同分布的;

接下来，通过以下方式定义极限过程 $\{S_{\frac{n}{N}}^{(N)}\}$

$$S_{\frac{k}{N}}^{(N)} = \frac{1}{\sqrt{N}} \sum_{i=1}^{k} X_i^{(N)};$$

缩放系数 $\frac{1}{\sqrt{N}}$ 是为了保持相同的方差。



$N \to \infty$: Normal distribution

Let us check the limiting distribution of $S_1^{(N)}$.

- Method 1: Brutal calculation (we leave as homework).
- Method 2: Consider the mgf of $S_1^{(N)}$, we have

$$M_{S_1^{(N)}}(t) = \mathbb{E}\left[e^{tS_1^{(N)}}\right] = \mathbb{E}\left[e^{\frac{t}{\sqrt{N}}\sum_{i=1}^{N} X_i^{(N)}}\right]$$
$$= \prod_{i=1}^{N} \mathbb{E}\left[e^{\frac{t}{\sqrt{N}} X_i^{(N)}}\right] = \left(\frac{1}{2}e^{\frac{t}{\sqrt{N}}} + \frac{1}{2}e^{-\frac{t}{\sqrt{N}}}\right)^N.$$

Using Taylor's expansion $e^x = 1 + x + \cdots + \frac{x^n}{n!} + \cdots$, we know that

$$\frac{1}{2}e^{\frac{t}{\sqrt{N}}} + \frac{1}{2}e^{-\frac{t}{\sqrt{N}}} = 1 + \frac{t^2}{2N} + \frac{t^4}{4!N^2} + \cdots$$

and then,

$$\lim_{N\to\infty} M_{S_1^{(N)}}(t) = \lim_{N\to\infty}\left(1 + \frac{t^2}{2N} + + \cdots\right)^N = e^{\frac{t^2}{2}}.$$

让我们检查 $S_1^{(N)}$ 的极限分布。

方法1：暴力计算（我们留作作业）。

方法2：考虑 $S_1^{(N)}$ 的矩母函数，我们有

$$M_{S_1^{(N)}}(t) = \mathbb{E}\left[e^{tS_1^{(N)}}\right] = \mathbb{E}\left[e^{\frac{t}{\sqrt{N}}\sum_{i=1}^{N} X_i^{(N)}}\right]$$
$$= \prod_{i=1}^{N} \mathbb{E}\left[e^{\frac{t}{\sqrt{N}} X_i^{(N)}}\right] = \left(\frac{1}{2}e^{\frac{t}{\sqrt{N}}} + \frac{1}{2}e^{-\frac{t}{\sqrt{N}}}\right)^N.$$

使用泰勒展开 $e^x = 1 + x + \cdots + \frac{x^n}{n!} + \cdots$，我们知道

$$\frac{1}{2}e^{\frac{t}{\sqrt{N}}} + \frac{1}{2}e^{-\frac{t}{\sqrt{N}}} = 1 + \frac{t^2}{2N} + \frac{t^4}{4!N^2} + \cdots$$

然后，

$$\lim_{N\to\infty} M_{S_1^{(N)}}(t) = \lim_{N\to\infty}\left(1 + \frac{t^2}{2N} + + \cdots\right)^N = e^{\frac{t^2}{2}}.$$

Normal distribution

This is (one of) the most important distributions in probability theory.

- Z follows the **standard normal distribution** if

$$M_Z(t) = e^{\frac{t^2}{2}};$$

- $\mathbb{E}[Z] = 0$, $\text{Var}(Z) = 1$ and pdf $f_Z(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}$.
- Consider the linear transformation of Z, that is $X = \mu + \sigma Z$, we call such distribution a **normal distribution (Gaussian distribution)** $\mathsf{N}(\mu, \sigma^2)$ if

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}},$$

and $\mathbb{E}[X] = \mu$, $\text{Var}(X) = \sigma^2$ and pdf $M_X(t) = e^{\frac{\mu t + \sigma^2 t^2}{2}}$.
  - Note that a normal distribution is uniquely determined by its mean μ and variance σ^2.

这是概率论中（一个）最重要的分布之一。

Z **如果遵循标准正态分布**

$$M_Z(t) = e^{\frac{t^2}{2}};$$

$\mathbb{E}[Z] = 0$, $\text{Var}(Z) = 1$ and 概率密度函数 $f_Z(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}$.

考虑 Z的线性变换，即 $X = \mu + \sigma Z$，如果这种分布是正态分布（高斯分布） $\mathsf{N}(\mu, \sigma^2)$

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}},$$

并且 $\mathbb{E}[X] = \mu$, $\text{Var}(X) = \sigma^2$ 以及概率密度函数 $M_X(t) = e^{\frac{\mu t + \sigma^2 t^2}{2}}$。

请注意，正态分布是由其均值 μ 和方差 σ^2唯一确定的。

Outline

1. Bernoulli process
   - Definition: counting success
   - Distributions associated with Bernoulli process
   - Law of large number
   - **Convergence of random variables*
2. Poisson process
   - Continuous arrivals and definition
   - Distributions associated with Poisson process
   - Superposition and thinning
3. Simple random walk and Brownian motion
   - Simple random walk and continuous limit
   - Brownian motion
   - Central limit theorem

概要

1. 伯努利过程
   - 定义：计数成功
   - 与伯努利过程相关的分布
   - 大数定律
   - *随机变量收敛
2. 泊松过程
   - 连续到达和定义
   - 与泊松过程相关的分布
   - 叠加和稀疏
3. 简单随机游走和布朗运动
   - 简单随机游走和连续极限
   - 布朗运动
   - 中心极限定理

Characterizing the limiting process

Similarly as in Poisson process, let us collect some useful properties of the limiting process $\{S_t^{\text{lim}}\}$ inherited from $\{S_t^{(N)}\}$:

- *independent increment*: In the discrete model, each of the steps $X_i^{(N)}$ is independent, and we can see that the increments from s to t, $S_t^{(N)} - S_s^{(N)}$ and $S_r^{(N)}$ depend on disjoint subsets of these variables. It follows immediately that

$$S_t^{(N)} - S_s^{(N)} \perp\!\!\!\perp \{S_r^{(N)}\}_{r \le s};$$

- *stationary increment*: Note that $S_t^{(N)} - S_s^{(N)}$ is the sum of $N(t-s)$ individual steps. As the steps in our model are i.i.d., this implies that $S_t^{(N)} - S_s^{(N)}$ has the same distribution as $S_{t-s}^{(N)}$ that is

$$S_t^{(N)} - S_s^{(N)} \overset{d}{=} S_{t-s}^{(N)}.$$

类似地，如在泊松过程中，让我们收集一些有用的性质，这些性质是极限过程 $\{S_t^{\text{lim}}\}$从 $\{S_t^{(N)}\}$继承的：

独立增量：在离散模型中，每个步骤$X_i^{(N)}$ 都是独立的，我们可以看到从 s 到t、 $S_t^{(N)} - S_s^{(N)}$和 $S_r^{(N)}$的增量依赖于这些变量的不相交子集。由此立即得出

$$S_t^{(N)} - S_s^{(N)} \perp\!\!\!\perp \{S_r^{(N)}\}_{r \le s};$$

平稳增量：请注意， $S_t^{(N)} - S_s^{(N)}$是$N(t-s)$ 个独立步骤的总和。由于我们模型中的步骤是独立同分布的，这意味着 $S_t^{(N)} - S_s^{(N)}$ 与 $S_{t-s}^{(N)}$ 具有相同的分布，即

$$S_t^{(N)} - S_s^{(N)} \overset{d}{=} S_{t-s}^{(N)}.$$

Brownian motion

The two properties described above, together with the distribution characterizations, completely determine the continuous limit arising from our modeling assumptions. We call this continuous limit process **Brownian motion**, typically denoted by $\{B_t\}_{t\geq 0}$.

- A continuous process $\{B_t\}_{t\geq 0}$ is called a **(standard) Brownian motion** if it satisfies the following properties: for all $s \leq t$,
  - $B_0 = 0$.
  - independent increment: $B_t - B_s \perp\!\!\!\perp \{B_r\}_{r\leq s}$;
  - stationary increments: $B_t - B_s \sim \mathrm{N}(0, t-s)$.

上述两种性质，连同分布特征，完全决定了由我们的建模假设产生的连续极限。我们称这个连续极限过程为布朗运动，通常用 $\{B_t\}_{t\geq 0}$ 表示。

一个连续过程 $\{B_t\}_{t\geq 0}$ 如果满足以下性质，则被称为（标准）布朗运动：对于所有 $s \leq t$，$B_0 = 0$。

独立增量： $B_t - B_s \perp\!\!\!\perp \{B_r\}_{r\leq s}$;
平稳增量： $B_t - B_s \sim \mathrm{N}(0, t-s)$.

Characterizing of Brownian motion

Let us consider the finite dimensional distribution of Brownian motion: $(B_{t_1}, B_{t_2}, \cdots, B_{t_n})$ for $t_1 < t_2 < \cdots < t_n$.

- Characterizing $(B_{t_1}, B_{t_2} - B_{t_1}, \cdots, B_{t_n} - B_{t_{n-1}})$ is more straightforward, we have

$$\begin{aligned}&f_{B_{t_1}, B_{t_2}-B_{t_1}, \cdots, B_{t_n}-B_{t_{n-1}}}(x_1, \Delta x_2, \cdots, \Delta x_n)\\ &\quad = f_{B_{t_1}}(x_1) f_{B_{t_2}-B_{t_1}}(\Delta x_2) \cdots f_{B_{t_n}-B_{t_{n-1}}}(\Delta x_n)\\ &\quad = \frac{1}{\sqrt{2\pi t_1}} e^{-\frac{x_1^2}{2t_1}} \frac{1}{\sqrt{2\pi(t_2-t_1)}} e^{-\frac{\Delta x_2^2}{2(t_2-t_1)}} \cdots \frac{1}{\sqrt{2\pi(t_n-t_{n-1})}} e^{-\frac{\Delta x_n^2}{2(t_n-t_{n-1})}}.\end{aligned}$$

- Using *transformation of random variables*, we can characterize the distribution of $(B_{t_1}, B_{t_2}, \cdots, B_{t_n})$ by

$$\begin{aligned}&f_{B_{t_1}, B_{t_2}, \cdots, B_{t_n}}(x_1, x_2, \cdots, x_n)\\ &= f_{B_{t_1}, B_{t_2}-B_{t_1}, \cdots, B_{t_n}-B_{t_{n-1}}}(x_1, x_2 - x_1, \cdots, x_n - x_{n-1})\\ &= \frac{1}{\sqrt{(2\pi)^n t_1 \cdots (t_n - t_{n-1})}} \exp\left(-\frac{x_1^2}{2t_1} - \frac{(x_2-x_1)^2}{2(t_2-t_1)} - \cdots - \frac{(x_n-x_{n-1})^2}{2(t_n-t_{n-1})}\right).\end{aligned}$$

让我们考虑布朗运动的有限维分布：
$(B_{t_1}, B_{t_2}, \cdots, B_{t_n})$ for $t_1 < t_2 < \cdots < t_n$.

描述 $(B_{t_1}, B_{t_2} - B_{t_1}, \cdots, B_{t_n} - B_{t_{n-1}})$ 的特征更加直接，我们有

$$\begin{aligned}&f_{B_{t_1}, B_{t_2}-B_{t_1}, \cdots, B_{t_n}-B_{t_{n-1}}}(x_1, \Delta x_2, \cdots, \Delta x_n)\\ &\quad = f_{B_{t_1}}(x_1) f_{B_{t_2}-B_{t_1}}(\Delta x_2) \cdots f_{B_{t_n}-B_{t_{n-1}}}(\Delta x_n)\\ &\quad = \frac{1}{\sqrt{2\pi t_1}} e^{-\frac{x_1^2}{2t_1}} \frac{1}{\sqrt{2\pi(t_2-t_1)}} e^{-\frac{\Delta x_2^2}{2(t_2-t_1)}} \cdots \frac{1}{\sqrt{2\pi(t_n-t_{n-1})}} e^{-\frac{\Delta x_n^2}{2(t_n-t_{n-1})}}.\end{aligned}$$

通过随机变量的变换，我们可以描述 $(B_{t_1}, B_{t_2}, \cdots, B_{t_n})$ 的分布。

$$\begin{aligned}&f_{B_{t_1}, B_{t_2}, \cdots, B_{t_n}}(x_1, x_2, \cdots, x_n)\\ &= f_{B_{t_1}, B_{t_2}-B_{t_1}, \cdots, B_{t_n}-B_{t_{n-1}}}(x_1, x_2 - x_1, \cdots, x_n - x_{n-1})\\ &= \frac{1}{\sqrt{(2\pi)^n t_1 \cdots (t_n - t_{n-1})}} \exp\left(-\frac{x_1^2}{2t_1} - \frac{(x_2-x_1)^2}{2(t_2-t_1)} - \cdots - \frac{(x_n-x_{n-1})^2}{2(t_n-t_{n-1})}\right).\end{aligned}$$

Transformation of random variables

For completeness, we present the transformation formula for random variables. Specifically, let X and Y be two d-dimensional random variables related by $Y = g(X)$ and $X = h(Y)$ where g and h are smooth functions.

- *If X is a discrete .v. with pmf f_X, then Y has pmf given by*

$$f_Y(y) = f_X\big(h(y)\big).$$

- *If X is a continuous r.v. with pmf f_X, then Y has pdf given by*

$$f_Y(y) = f_X\big(h(y)\big)\big|\det\big(J_h(y)\big)\big|$$

where $J_h(y)$ denotes the Jacobian matrix of h at y:

$$J_h(y) = \begin{pmatrix} \frac{\partial h_1}{y_1} & \dots & \frac{\partial h_1}{y_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial h_n}{y_1} & \dots & \frac{\partial h_n}{y_n} \end{pmatrix}.$$

为了完整起见，我们展示了随机变量的变换公式。具体来说，设 X 和 Y 是两个由 $Y = g(X)$ 和 $X = h(Y)$ 关联的 d-维随机变量，其中 g 和 h 是光滑函数。

如果 X 是一个离散随机变量，其概率质量函数为 f_X，那么 Y 的概率质量函数为

$$f_Y(y) = f_X\big(h(y)\big).$$

如果 X 是一个连续随机变量，其概率密度函数为 f_X，那么 Y 的概率密度函数为

$$f_Y(y) = f_X\big(h(y)\big)\big|\det\big(J_h(y)\big)\big|$$

where $J_h(y)$表示 h在 y处的雅可比矩阵：

$$J_h(y) = \begin{pmatrix} \frac{\partial h_1}{y_1} & \dots & \frac{\partial h_1}{y_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial h_n}{y_1} & \dots & \frac{\partial h_n}{y_n} \end{pmatrix}.$$

Joint normal distribution

- We now return to the distribution of $(B_{t_1}, \cdots, B_{t_n})$. This distribution admits a simple characterization: it arises as linear transformation of a mutually independent normal r.v.'s $(B_{t_1}, B_{t_2} - B_{t_1}, \cdots, B_{t_n} - B_{t_{n-1}})$.
- Note that, any linear combination of $B_{t_1}, \cdots, B_{t_n}$ is a linear combination of mutually independent normal r.v.'s, and we can show that it must be normally distribution (leave as homework), that is

$$\sum_{i=1}^{n} a_i B_{t_i} \text{ is normally distributed for all } a_1, \cdots, a_n \in \mathbb{R}.$$

- More generally, we introduce the notion of *jointly normal (jointly Gaussian)* through this property, that is $X_1, X_2, \cdots, X_n$ are **jointly normal (jointly Gaussian)** if every linear combination $\sum_{i=1}^{n} a_i X_i$ is normally distributed for every $a_1, \cdots, a_n \in \mathbb{R}$.

我们现在回到 $(B_{t_1} \cdots B_{t_n})$，，的分布。这种分布有一个简单的特征：它作为一组相互独立的正态随机变量 $(B_{t_1}, B_{t_2} - B_{t_1}, \cdots, B_{t_n} - B_{t_{n-1}})$.的线性变换而产生。

注意， $B_{t_1} \cdots B_{t_n}$ 的任何线性组合是相互独立正态随机变量的线性组合，并且我们可以证明它必须是正态分布（留作作业），即

$$\sum_{i=1}^{n} a_i B_{t_i} \text{ is normally distributed for all } a_1, \cdots, a_n \in \mathbb{R}.$$

更一般地，我们通过这一性质引入联合正态（联合高斯）的概念，即 $X_1\ X_2 \cdots X_n$，，如果对于每一个线性组合 $\sum_{i=1}^{n} a_i X_i$ 在每一个 $a_1 \cdots a_n \in \mathbb{R}$，，上都是正态分布的，则是联合正态（联合高斯）的。

Property of jointly normal distribution

Suppose $\boldsymbol{X} = (X_1, \cdots, X_n)$ is jointly normal distribution.

- Further assume that $\boldsymbol{X}$ has mean $\boldsymbol{\mu} = (\mu_1, \cdots, \mu_n)$ and covariance matrix $\boldsymbol{\Sigma} = (\Sigma_{ij})$, that is

$$\mu_i = \mathbb{E}[X_i], \text{ and } \Sigma_{ij} = \mathrm{Cov}(X_i, X_j) = \mathbb{E}[X_i X_j] - \mathbb{E}[X_i]\mathbb{E}[X_j].$$

- Characterizing $\boldsymbol{a}^\top \boldsymbol{X}$ for $\boldsymbol{a} = (a_1, \cdots, a_n)$ by its mean and variance:
  - $\mathbb{E}\left[\boldsymbol{a}^\top \boldsymbol{X}\right] = \boldsymbol{a}^\top \mathbb{E}[\boldsymbol{X}] = \boldsymbol{a}^\top \boldsymbol{\mu}$.
  - $\mathrm{Var}(\boldsymbol{a}^\top \boldsymbol{X}) = \boldsymbol{a}^\top \mathrm{Cov}(\boldsymbol{X}) \boldsymbol{a} = \boldsymbol{a}^\top \boldsymbol{\Sigma} \boldsymbol{a}$.
  - Together with the fact that $\boldsymbol{a}^\top \boldsymbol{X}$ is normally distributed, we know that

$$\boldsymbol{a}^\top \boldsymbol{X} \sim \mathrm{N}\left(\boldsymbol{a}^\top \boldsymbol{\mu}, \boldsymbol{a}^\top \boldsymbol{\Sigma} \boldsymbol{a}\right)$$

- We denote such random variables by $\mathrm{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ and its pdf is

$$f_{\boldsymbol{X}}(\boldsymbol{x}) = \frac{1}{(2\pi)^{\frac{d}{2}} |\det(\boldsymbol{\Sigma})|} \exp\left\{ -\frac{1}{2} (\boldsymbol{x} - \boldsymbol{\mu})^\top \Sigma^{-1} (\boldsymbol{x} - \boldsymbol{\mu}) \right\}.$$

假设 $\boldsymbol{X} = (X_1, \cdots, X_n)$ 是联合正态分布。

进一步假设 $\boldsymbol{X}$ 的均值为 $\boldsymbol{\mu} = (\mu_1, \cdots, \mu_n)$ 且协方差为矩阵 $\boldsymbol{\Sigma} = (\Sigma_{ij})$，即

$$\mu_i = \mathbb{E}[X_i] \text{ and } \Sigma_{ij} = \mathrm{Cov}(X_i, X_j) = \mathbb{E}[X_i X_j] - \mathbb{E}[X_i]\mathbb{E}[X_j].,$$

通过均值和方差来描述 $\boldsymbol{a}^\top \boldsymbol{X}$ 的特征：$\boldsymbol{a} = (a_1, \cdots, a_n)$

$\mathbb{E}\left[\boldsymbol{a}^\top \boldsymbol{X}\right] = \boldsymbol{a}^\top \mathbb{E}[\boldsymbol{X}] = \boldsymbol{a}^\top \boldsymbol{\mu}$.

Var $(\boldsymbol{a}^\top \boldsymbol{X}) = \boldsymbol{a}^\top$Cov$(\boldsymbol{X})\boldsymbol{a} = \boldsymbol{a}^\top \boldsymbol{\Sigma} \boldsymbol{a}$. 结合 $\boldsymbol{a}^\top \boldsymbol{X}$ 正态分布的事实，我们

知道

$$\boldsymbol{a}^\top \boldsymbol{X} \sim \mathrm{N}\left(\boldsymbol{a}^\top \boldsymbol{\mu}, \boldsymbol{a}^\top \boldsymbol{\Sigma} \boldsymbol{a}\right)$$

我们用 $\mathrm{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ 表示此类随机变量，其概率密度函数为

$$f_{\boldsymbol{X}}(\boldsymbol{x}) = \frac{1}{(2\pi)^{\frac{d}{2}} |\det(\boldsymbol{\Sigma})|} \exp\left\{ -\frac{1}{2} (\boldsymbol{x} - \boldsymbol{\mu})^\top \Sigma^{-1} (\boldsymbol{x} - \boldsymbol{\mu}) \right\}.$$

Characterizing of Brownian motion

Coming back to the finite dimensional distribution of Brownian motion: $(B_{t_1}, B_{t_2}, \cdots, B_{t_n})$ for $t_1 < t_2 < \cdots < t_n$:

- Following the introduction of jointly normal distribution, we conclude that

$$(B_{t_1}, B_{t_2}, \cdots, B_{t_n}) \sim \mathsf{N}(\mathbf{0}, \mathbf{\Sigma}),$$

  where $\mathbf{\Sigma} = (\Sigma_{i,j})$ and $\Sigma_{i,j} = \mathrm{Cov}(B_{t_i}, B_{t_j})$.

- Calculating $\mathrm{Cov}(B_{t_i}, B_{t_j})$, for $t_i < t_j$, we have

$$\begin{aligned}\mathrm{Cov}(B_{t_i}, B_{t_j}) &= \mathbb{E}\big[B_{t_i}B_{t_j}\big] - \mathbb{E}[B_{t_i}]\mathbb{E}[B_{t_j}] = \mathbb{E}\big[B_{t_i}B_{t_j}\big] \\ &= \mathbb{E}\big[B_{t_i}^2\big] + \mathbb{E}\big[B_{t_i}\big(B_{t_j} - B_{t_i}\big)\big] = t_i.\end{aligned}$$

- To summarize, we have $\Sigma_{ij} = t_i$ for $i < j$, and

$$(B_{t_1}, B_{t_2}, \cdots, B_{t_n}) \sim \mathsf{N}(\mathbf{0}, \mathbf{\Sigma}).$$

回到布朗运动的有限维分布：
$(B_{t_1}, B_{t_2}, \cdots, B_{t_n})$ for $t_1 < t_2 < \cdots < t_n$:

在联合正态分布的介绍之后，我们得出结论

$$(B_{t_1}, B_{t_2}, \cdots, B_{t_n}) \sim \mathsf{N}(\mathbf{0}, \mathbf{\Sigma}),$$

其中 $\mathbf{\Sigma} = (\Sigma_{i,j})$ 和 $\Sigma_{i,j} = \mathrm{Cov}(B_{t_i}, B_{t_j})$.

计算 $\mathrm{Cov}(B_{t_i}, B_{t_j})$，对于 $t_i < t_j$，我们有

$$\begin{aligned}\mathrm{Cov}(B_{t_i}, B_{t_j}) &= \mathbb{E}\big[B_{t_i}B_{t_j}\big] - \mathbb{E}[B_{t_i}]\mathbb{E}[B_{t_j}] = \mathbb{E}\big[B_{t_i}B_{t_j}\big] \\ &= \mathbb{E}\big[B_{t_i}^2\big] + \mathbb{E}\big[B_{t_i}\big(B_{t_j} - B_{t_i}\big)\big] = t_i.\end{aligned}$$

总而言之，我们为 $i < j$准备了 $\Sigma_{ij} = t_i$ ，并且

$$(B_{t_1}, B_{t_2}, \cdots, B_{t_n}) \sim \mathsf{N}(\mathbf{0}, \mathbf{\Sigma}).$$

Outline

1. Bernoulli process
   - Definition: counting success
   - Distributions associated with Bernoulli process
   - Law of large number
   - **Convergence of random variables*
2. Poisson process
   - Continuous arrivals and definition
   - Distributions associated with Poisson process
   - Superposition and thinning
3. Simple random walk and Brownian motion
   - Simple random walk and continuous limit
   - Brownian motion
   - Central limit theorem

概要

1. 伯努利过程
   - 定义：计数成功
   - 与伯努利过程相关的分布
   - 大数定律
   - *随机变量收敛
2. 泊松过程
   - 连续到达和定义
   - 与泊松过程相关的分布
   - 叠加和稀疏
3. 简单随机游走和布朗运动
   - 简单随机游走和连续极限
   - 布朗运动
   - 中心极限定理

Revisit the limiting distribution

In this subsection, we will explore why the normal distribution is among the most important probability distributions. Let us begin by recalling how the standard normal distribution was introduced:

- The standard normal distribution $\mathrm{N}(0,1)$ is defined after the following scaling limit:

$$S_1^{(N)} = \frac{1}{\sqrt{N}} \sum_{i=1}^{N} X_i^{(N)} \to \mathrm{N}(0,1) \quad \text{as } n \to \infty.$$

- The key step is to calculate the limiting mgf,

$$M_{S_1^{(N)}}(t) = \left(\mathbb{E}\left[e^{\frac{X_1^{(N)} t}{\sqrt{N}}}\right]\right)^N = \left(M_{X_1^{(N)}}\left(\frac{t}{\sqrt{N}}\right)\right)^N \to e^{\frac{t^2}{2}}.$$

- The final limit depends on our specific choice of distribution for $X_1^{(N)}$. What happens if we consider more general distributions?

在本小节中，我们将探讨为什么正态分布是其中最重要的概率分布之一。让我们从回顾标准正态分布是如何被引入开始:

标准正态分布 N(0,1) 是在以下缩放极限之后定义的:

$$S_1^{(N)} = \frac{1}{\sqrt{N}} \sum_{i=1}^{N} X_i^{(N)} \to \mathrm{N}(0,1) \quad \text{as } n \to \infty.$$

关键步骤是计算极限矩母函数,

$$M_{S_1^{(N)}}(t) = \left(\mathbb{E}\left[e^{\frac{X_1^{(N)} t}{\sqrt{N}}}\right]\right)^N = \left(M_{X_1^{(N)}}\left(\frac{t}{\sqrt{N}}\right)\right)^N \to e^{\frac{t^2}{2}}.$$

最终极限取决于我们对$X_1^{(N)}$ 的特定分布选择。如果我们考虑更一般的分布会怎样?

More general distribution

Now we assume that $X_1, \cdots, X_n, \cdots$ has a more general distribution and still set

$$S^{(N)} := \frac{1}{\sqrt{N}} \sum_{i=1}^{N} X_i.$$

- Recall the law of large numbers, which tells us that

$$\frac{S^{(N)}}{\sqrt{N}} = \frac{1}{N} \sum_{i=1}^{N} X_i \to \mathbb{E}[X_1],$$

  Therefore, if $\mathbb{E}[X_1^{(N)}] \neq 0$, it is clear that $S^{(N)} \to \infty$.
- Without loss of generality, we can assume that $\mathbb{E}[X_1] = 0$, and $\mathrm{Var}(X_1) = \mathbb{E}[X_1^2] = \sigma^2$ (Otherwise, we set $\widetilde{X}_i = X_i - \mathbb{E}[X_i]$.)

现在我们假设 $X_1, \cdots, X_n, \cdots$ 具有更一般的分布，仍然设置

$$S^{(N)} := \frac{1}{\sqrt{N}} \sum_{i=1}^{N} X_i.$$

回顾大数定律，它告诉我们

$$\frac{S^{(N)}}{\sqrt{N}} = \frac{1}{N} \sum_{i=1}^{N} X_i \to \mathbb{E}[X_1],$$

因此，如果 $\mathbb{E}[X_1^{(N)}] \neq 0$，那么显然 $S^{(N)} \to \infty$。

不失一般性，我们可以假设 $\mathbb{E}[X_1] = 0$，和 $\mathrm{Var}(X_1) = \mathbb{E}[X_1^2] = \sigma^2$（否则，我们设置 $\widetilde{X}_i = X_i - \mathbb{E}[X_i]$.)）

More general distribution continued

- Now calculating the mgf of X and noting that

$$M_X(t) = \mathbb{E}[e^{tX}] = 1 + t \cdot \mathbb{E}[X] + \cdots + \frac{t^n}{n!}\mathbb{E}[X^n] + \cdots.$$

- Plugging back into the mgf of $S^{(N)}$, we have

$$M_{S^{(N)}}(t) = \left(M_{X_1}\left(\frac{t}{\sqrt{N}}\right)\right)^N = \left(1 + \frac{t^2}{2N}\mathbb{E}[X_1^2] + \sum_{n=3}^{\infty} \frac{t^n}{n!(N)^{\frac{n}{2}}}\mathbb{E}[X_1^n]\right)^N$$
$$\to e^{\frac{t^2\sigma^2}{2}} \text{ as } N \to \infty.$$

- Recall that the mgf of normal distribution, we know that the limiting distribution of $S^{(N)}$ should be $\mathrm{N}(0, \sigma^2)$. *Remarkably, this holds regardless of the distribution of X_1!*

现在计算 X 的矩母函数并注意到

$$M_X(t) = \mathbb{E}[e^{tX}] = 1 + t \cdot \mathbb{E}[X] + \cdots + \frac{t^n}{n!}\mathbb{E}[X^n] + \cdots.$$

将公式代入 $S^{(N)}$的矩母函数，我们得到

$$M_{S^{(N)}}(t) = \left(M_{X_1}\left(\frac{t}{\sqrt{N}}\right)\right)^N = \left(1 + \frac{t^2}{2N}\mathbb{E}[X_1^2] + \sum_{n=3}^{\infty} \frac{t^n}{n!(N)^{\frac{n}{2}}}\mathbb{E}[X_1^n]\right)^N$$
$$\to e^{\frac{t^2\sigma^2}{2}} \text{ as } N \to \infty.$$

回想一下正态分布的矩母函数，我们知道 $S^{(N)}$ 的极限分布应该是 $\mathrm{N}(0, \sigma^2)$。值得注意的是，无论 X_1的分布如何，这一点都成立！

Central limit theorem

The precise statement of **central limit theorem** is as follows

- If $\{X_k\}$ are i.i.d. r.v.'s with $\mathbb{E}[X_k] = 0$ and $\mathrm{Var}(X_k) = \sigma^2$, then

$$\frac{X_1 + X_2 + \cdots + X_N}{\sqrt{N}} \to \mathrm{N}(0, \sigma^2) \quad \text{as } N \to \infty.$$

- More generally, let $\bar{X} = \frac{1}{N}\sum_{i=1}^{N} X_i$ be the mean of a random sample $X_1, .., X_n$ from a distribution with finite mean μ and a finite positive variance σ^2, then

$$W = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \to \mathrm{N}(0, 1), \text{ as } n \to \infty.$$

  - Note that we do not need to know exactly the underlying distribution of X is.
  - Roughly speaking, $\bar{X} \sim \mathrm{N}(\mu, \frac{\sigma^2}{n})$ when n is large enough, which is exactly the same as the normal samples (i.e. $X_1 \sim \mathrm{N}(\mu, \sigma^2)$).

中心极限定理的精确表述如下

如果 $\{X_k\}$ 是独立同分布的随机变量，且均值为 $\mathbb{E}[X_k] = 0$ ，方差为 $\mathrm{Var}(X_k) = \sigma^2$ ，则

$$\frac{X_1 + X_2 + \cdots + X_N}{\sqrt{N}} \to \mathrm{N}(0, \sigma^2) \quad \text{as } N \to \infty.$$

更一般地，设 $\bar{X} = \frac{1}{N}\sum_{i=1}^{N} X_i$ 是从具有有限均值 μ 和有限正方差的分布中抽取的随机样本 X_1 .. X_n , , 的均值，则

$$W = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \to \mathrm{N}(0, 1), \text{ as } n \to \infty.$$

注意，我们不需要知道 X 的确切分布。

简单来说， $\bar{X} \sim \mathrm{N}(\mu, \frac{\sigma^2}{n})$ 当 n 足够大时，这正好与正态样本（即 $X_1 \sim \mathrm{N}(\mu, \sigma^2)$ ）相同。

Summary 1

1. Bernoulli process

- Definition: $\{X_i\}_{i=1,2,..}$ be a sequence of i.i.d. Bern(p) r.v.'s.
- Corresponding distributions:
  1. Bernoulli r.v. Bern(p): X_i,
     $\mathbb{E}[X_i] = p$, Var$(X_i) = p(1-p)$, $M_{X_i}(t) = pe^t + 1 - p$;
  2. Binomial r.v. B(n, p): $S_n := X_1 + \cdots X_n$;
     $\mathbb{E}[S_n] = np$, Var$(S_n) = np(1-p)$, $M_{S_n}(t) = (pe^t + 1 - p)^n$;
  3. Geometric r.v. Geo(p): $T_1 := \min\{n : S_n = 1\}$,
     $\mathbb{E}[T_1] = \frac{1}{p}$, Var$(T_1) = \frac{1-p}{p}$, $M_{T_1}(t) = \frac{pt}{1-(1-p)e^t}$;
  4. Negative Binomial r.v. NB(r, p): $T_r := \min\{n : S_n = 1\}$,
     $\mathbb{E}[T_r] = \frac{r}{p}$, Var$(T_1) = \frac{r(1-p)}{p}$, $M_{T_1}(t) = (\frac{pt}{1-(1-p)e^t})^r$.
     In particular, T_1, $T_2 - T_1$, $\cdots$, $T_n - T_{n-1}$ are i.i.d. r.v.'s.
- Law of large number: $\lim_{n\to\infty} \frac{S_n}{n} = p$. More generally, if X_i's are i.i.d., we have $\lim_{n\to\infty} \frac{1}{n}\sum_{i=1}^{n} X_i = \mathbb{E}[X_1]$.
  - Proof sketch: $\mathbb{E}[\frac{S_n}{n}] = p$ and Var$(\frac{S_n}{n}) = \frac{p(1-p)}{n}$.

1. 伯努利过程

定义： $\{X_i\}_{i=1,2,..}$ 是一个 i.i.d. Bern(p) 随机变量序列。
对应的分布：

Bernoulli r.v. Bern(p): X_i,
$\mathbb{E}[X_i] = p$, Var$(X_i) = p(1-p)$, $M_{X_i}(t) = pe^t + 1 - p$;
Binomial r.v. B(n, p): $S_n := X_1 + \cdots X_n$;
$\mathbb{E}[S_n] = np$, Var$(S_n) = np(1-p)$, $M_{S_n}(t) = (pe^t + 1 - p)^n$;
Geometric r.v. Geo(p): $T_1 := \min\{n : S_n = 1\}$,
$\mathbb{E}[T_1] = \frac{1}{p}$, Var$(T_1) = \frac{1-p}{p}$, $M_{T_1}(t) = \frac{pt}{(1-p)e^t}$;
负二项分布 r.v. NB r, p :(T_r : = min n : $S_n = 1\}$,
$\mathbb{E}[T_r] = \frac{r}{p}$, Var$(T_1) = \frac{r(1-p)}{p}$, $M_{T_1}(t) = (\frac{pt}{1-(1-p)e^t})^r$.
In particular, T_1, $T_2 - T_1$, $\cdots$, $T_n - T_{n-1}$ are i.i.d. r.v.'s.

大数定律： $\lim_{n\to\infty} \frac{S_n}{n} = p$。更一般地，如果 X_i’ s 因为是独立同分布的，我们有 $\lim_{n\to\infty} \frac{1}{n}\sum_{i=1}^{n} X_i = \mathbb{E}[X_1]$.

证明思路： $\mathbb{E}[\frac{S_n}{n}] = p$ 和 Var$(\frac{S_n}{n}) = \frac{p(1-p)}{n}$。

Summary 2

1. Bernoulli process (continued)
   - Convergence of r.v.'s
     - Almost surely convergence $X_n \xrightarrow{\text{a.s.}} X$: $\mathbb{P}(X_n \to X) = 1$;
     - Converge in probability $X_n \xrightarrow{\text{p}} X$: $\mathbb{P}(|X_n - X| \geq \epsilon) \to 0$;
     - Converge in L^p $X_n \xrightarrow{L^p} X$: $\mathbb{E}[|X_n - X|^p] \to 0$;
     - Converge in distribution: $X_n \xrightarrow{d} X$: $F_n(x) \to F_n(x)$;
     - Weak LLN: Suppose $\{X_i\}$ are i.i.d. with finite mean, then for all $\epsilon > 0$, $\lim_{n\to\infty} \mathbb{P}\left(\left|\frac{\sum_{i=1}^n X_i}{n} - \mathbb{E}[X_1]\right| \geq \epsilon\right) = 0$.
2. Poisson process
   - Continuous time limit: for $m = 1, 2, \ldots$, consider the process:

$$S^{(m)}_{\frac{k}{m}} = \sum_{i=1}^{k} X_i^{(m)}, \text{ where } X_i^{(m)} \sim \text{Bern}\left(\frac{p}{m}\right).$$

   - $\{S_t^{\lim}\} := \lim_{m\to\infty} \{S_t^{(m)}\}$ is the desired Poisson process;
   - $\lim_{m\to\infty} S_1^{(m)} = S_1^{\lim} \sim \text{Pois}(p)$.

1. 伯努利过程（续）

Convergence of r.v.'s 几乎处处收敛 $X_n \xrightarrow{\text{a.s.}} X$: $\mathbb{P}(X_n \to X) = 1$;

依概率收敛 X_n p→ X: $\mathbb{P}(|X_n - X| \geq \epsilon) \to 0$;

依 L^p $X_n \xrightarrow{L^p} X$收敛: $\mathbb{E}[|X_n - X|^p] \to 0$;

依分布收敛: $X_n \xrightarrow{d} X$: $F_n(x) \to F_n(x)$;

弱大数定律：假设 $\{X_i\}$ 是独立同分布且具有有限均值，那么对于所有 $\epsilon > 0$ $\lim_{n\to\infty} \mathbb{P}\left(\left|\frac{\sum_{i=1}^n X_i}{n} - \mathbb{E}[X_1]\right| \geq \epsilon\right) = 0.$,

2. 泊松过程

连续时间极限：对于 $m = 1$ 2 ..., , , 考虑过程：

$$S^{(m)}_{\frac{k}{m}} = \sum_{i=1}^{k} X_i^{(m)}, \text{ where } X_i^{(m)} \sim \text{Bern}\left(\frac{p}{m}\right).$$

$\{S_t^{\lim}\} := \lim_{m\to\infty} \{S_t^{(m)}\}$ 是期望的泊松过程;

$\lim_{m\to\infty} S_1^{(m)} = S_1^{\lim} \sim \text{Pois}(p)$.

Summary 3

2. Poisson process (continued)

- Definition: A counting process $\{N_t\}_{t\geq 0}$ is called **a Poisson process with rate** $\lambda > 0$ if for all $s \leq t$,
  - $N_0 = 0$.
  - independent increment: $N_t - N_s \perp\!\!\!\perp \{N_r\}_{r\leq s}$;
  - stationary increments: $N_t - N_s \sim \text{Pois}(\lambda(t-s))$.
- Corresponding distributions:
  1. Poisson r.v. Pois(λ): S_1,
     $\mathbb{E}[S_1] = \lambda$, Var$(S_1) = \lambda$, $M_{S_1}(t) = e^{\lambda(e^t-1)}$;
  2. Exponential r.v. Exp(λ): $T_1 := \min\{t : S_t = 1\}$,
     $\mathbb{E}[T_1] = \frac{1}{\lambda}$, Var$(T_1) = \frac{1}{\lambda^2}$, $M_{T_1}(t) = \frac{\lambda}{\lambda - t}$;
  3. Gamma r.v. Gamma(r, p): $T_r := \min\{r : S_t = r\}$,
     $\mathbb{E}[T_r] = \frac{r}{\lambda}$, Var$(T_1) = \frac{r}{\lambda^2}$, $M_{T_1}(t) = (\frac{\lambda}{\lambda - t})^r$.
     In particular, T_1, $T_2 - T_1$, $\cdots$, $T_n - T_{n-1}$ are i.i.d. r.v.'s.
     Furthermore, r can be extended to non-integer values.

2. 泊松过程（续）

定义：一个计数过程 $\{N_t\}_{t\geq 0}$ 被称为具有速率 $\lambda > 0$ 的泊松过程，如果对于所有 $s \leq t$,

$N_0 = 0$.
独立增量： $N_t - N_s \perp\!\!\!\perp \{N_r\}_{r\leq s}$;
平稳增量： $N_t - N_s \sim \text{Pois}(\lambda(t-s))$。

相应的分布：

Poisson r.v. Pois(λ): S_1,
$\mathbb{E}[S_1] = \lambda$, Var$(S_1) = \lambda$, $M_{S_1}(t) = e^{\lambda(e^t-1)}$;
Exponential r.v. Exp(λ): $T_1 := \min\{t : S_t = 1\}$,
$\mathbb{E}[T_1] = \frac{1}{\lambda}$, Var$(T_1) = \frac{1}{\lambda^2}$, $M_{T_1}(t) = \frac{\lambda}{\lambda - t}$;
Gamma r.v. Gamma(r, p): $T_r := \min\{r : S_t = r\}$,
$\mathbb{E}[T_r] = \frac{r}{\lambda}$, Var$(T_1) = \frac{r}{\lambda^2}$, $M_{T_1}(t) = (\frac{\lambda}{\lambda - t})^r$.
In particular, T_1, $T_2 - T_1$, $\cdots$, $T_n - T_{n-1}$ are i.i.d. r.v.'s.
Furthermore, r can be extended to non-integer values.

Summary 4

2. Poisson process (continued)
   - Superposition and thinning.
     - Superposition principle: If $\{N^{(1)}\}_t, \cdots, \{N^{(k)}\}_t$ are independent Poisson processes with rates $\mu_1, \cdots, \mu_k$, then $\{N^{(1)} + \cdots + N^{(k)}\}_t$ is a Poisson process with rate $\mu_1 + \cdots + \mu_k$.
     - Thinning principle: If N_t is a Poisson process with rate μ, and if each arrival is independently assigned one of k labels with probabilities $p_1, \cdots, p_k$, then the counting processes $\{N_t^{(1)}\}, \cdots, \{N_t^{(k)}\}$ that count the arrivals of each type are independent Poisson processes with rates $p_1\mu, \cdots, p_k\mu$.
3. Simple random walk (SRW) and Brownian motion
   - SRW definition: $\{S_n\}_{n=0,1,2,\dots}$ with $S_n = S_0 + X_1 + \cdots + X_n$ where $\{X_i\}$ are i.i.d. r.v.'s taking values on ± 1.
     - We focus on symmetric simple random walks where $\mathbb{P}(X_1 = 1) = \mathbb{P}(X_1 = -1) = \frac{1}{2}$.

2. 泊松过程（续）

叠加和稀疏

叠加原理：如果 $\{N^{(1)}\}_t$ $\cdots$ $\{N^{(k)}\}_t$，，是具有速率 μ_1 $\cdots$ μ_k，，的独立泊松过程，则 $\{N^{(1)} + \cdots + N^{(k)}\}_t$ 是具有速率 $\mu_1 + \cdots + \mu_k$ 的泊松过程。

稀疏原理：如果 N_t 是具有速率 μ 的泊松过程，并且每个到达独立地被分配到 k 个标签之一，其概率为 $p_1, \cdots, p_k$，则计数过程 $\{N_t^{(1)}\}$ $\cdots$ $\{N_t^{(k)}\}$，，计算每种类型的到达次数是独立的泊松过程，其速率为 $p_1\mu$ $\cdots$ $p_k\mu$，，。

3. 简单随机游走（SRW）和布朗运动 SRW 定义：$\{S_n\}_{n=0,1,2,\cdots}$，其中 $S_n = S_0 + X_1 + \cdots + X_n$，$\{X_i\}$ 是独立同分布的随机变量，取值于 ± 1。

我们关注对称的简单随机游走，其中 $\mathbb{P}(X_1 = 1) = \mathbb{P}(X_1 = -1) = \frac{1}{2}$.

Summary 5

3. Simple random walk (SRW) and Brownian motion (continued)
   - Continuous time limit: for $m = 1, 2, ...$, consider the process:

$$S^{(N)}_{\frac{k}{N}} = \frac{1}{\sqrt{N}} \sum_{i=1}^{k} X_i^{(N)}, \text{ where } X_i^{(N)} \text{ are i.i.d.}$$

   with $\mathbb{P}(X_i^N = 1) = \mathbb{P}(X_i^N = -1) = \frac{1}{2}$.
     - $\{S_t^{\lim}\} := \lim_{N\to\infty}\{S_t^{(N)}\}$ is the desired Brownian motion;
     - $\lim_{N\to\infty} S_1^{(N)} = S_1^{\lim} \sim \mathrm{N}(0,1)$.
   - Definition: A continuous process $\{B_t\}_{t\geq 0}$ is called a **(standard) Brownian motion** if for all $s \leq t$,
     - $B_0 = 0$.
     - independent increment: $B_t - B_s \perp\!\!\!\perp \{B_r\}_{r\leq s}$;
     - stationary increments: $B_t - B_s \sim \mathrm{N}(0, t-s)$.

3. 简单随机游走（SRW）和布朗运动（续）连续时间极限：对于 $m = 1\ 2\ ...,$ ，， 考虑过程：

$$S^{(N)}_{\frac{k}{N}} = \frac{1}{\sqrt{N}} \sum_{i=1}^{k} X_i^{(N)}, \text{ where } X_i^{(N)} \text{ are i.i.d.}$$

with $\mathbb{P}(X_i^N = 1) = \mathbb{P}(X_i^N = -1) = \frac{1}{2}$.

$\{S_t^{\lim}\} := \lim_{N\to\infty}\{S_t^{(N)}\}$ 是期望的布朗运动；

$\lim_{N\to\infty} S_1^{(N)} = S_1^{\lim} \sim \mathrm{N}(0,1)$。

定义：一个连续过程 $\{B_t\}_{t\geq 0}$ 被称为（标准）布朗运动，如果对于所有 $s \leq t$,

$B_0 = 0$.

独立增量： $B_t - B_s \perp\!\!\!\perp \{B_r\}_{r\leq s}$;，平稳增量： $B_t - B_s \sim \mathrm{N}(0, t-s)$。

74 / 76

Summary 6

3. Simple random walk (SRW) and Brownian motion (continued)
   - Normal distribution
     - Normal r.v. $\mathrm{N}(\mu, \sigma^2)$ X, $f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$,

       $\mathbb{E}[X] = \mu$, $\mathrm{Var}(X) = \sigma^2$, $M_X(t) = e^{\mu t + \frac{\sigma^2 t^2}{2}}$.
   - Joint Normal distribution
     - $\boldsymbol{X} = (X_1, X_2, \cdots, X_n)$ are **jointly normal (Gaussian)** if $\sum_{i=1}^n a_i X_i$ is normally distributed for all $a_1, \cdots, a_n \in \mathbb{R}$.
     - $\boldsymbol{X} \sim \mathrm{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ with $\mathbb{E}[\boldsymbol{X}] = \boldsymbol{\mu}$ and Covariance matrix is Σ.
   - Characterization of Brownian motion
     - For $t_1 \le t_2 \le \cdots \le t_n$, we have

$$(B_{t_1}, B_{t_2}, \cdots, B_{t_n}) \sim \mathrm{N}\left(\mathbf{0}, \begin{pmatrix} t_1 & t_1 & \cdots & t_1 \\ t_1 & t_2 & \cdots & t_2 \\ \vdots & \vdots & \ddots & \vdots \\ t_1 & t_2 & \cdots & t_n \end{pmatrix}\right).$$

3. 简单随机游走（SRW）和布朗运动（续）

正态分布

正态随机变量 $\mathrm{N}(\mu, \sigma^2)$ X, $f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$,

[$\mathbb{E}$ X =( ) 联合正态分 Var X $= \sigma^2$, $M_X(t) = e^{\mu t + \frac{\sigma^2 t^2}{2}}$.

布

$\boldsymbol{X} = (X_1, X_2, \cdots, X_n)$ 是联合正态（高斯）分布的，如果对于所有 a_1 $\cdots$ $a_n \in \mathbb{R}$.，, 是正态 $\sum_{i=1}^n a_i X_i$
分布的

$\boldsymbol{X} \sim \mathrm{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ 与 $\mathbb{E}[\boldsymbol{X}] = \boldsymbol{\mu}$，协方差矩阵是 Σ。

布朗运动的特性

对于 $t_1 \le t_2 \le \cdots \le t_n$，我们有

$$(B_{t_1}, B_{t_2}, \cdots, B_{t_n}) \sim \mathrm{N}\left(\mathbf{0}, \begin{pmatrix} t_1 & t_1 & \cdots & t_1 \\ t_1 & t_2 & \cdots & t_2 \\ \vdots & \vdots & \ddots & \vdots \\ t_1 & t_2 & \cdots & t_n \end{pmatrix}\right).$$

Summary 7

3. Simple random walk (SRW) and Brownian motion (continued)
    - Central Limit theorem
        - Let $\bar{X} = \frac{1}{N}\sum_{i=1}^{N} X_i$ be the mean of a random sample $X_1, .., X_n$ from a distribution with finite mean μ and a finite positive variance σ^2, then

$$W = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \to \mathrm{N}(0,1), \text{ as } n \to \infty.$$

        - We do not need to know the underlying distribution of X_i is.
        - $\bar{X} \sim \mathrm{N}(\mu, \frac{\sigma^2}{n})$ when n is large enough, which is exactly the same as the normal samples (i.e. $X_1 \sim \mathrm{N}(\mu, \sigma^2)$).

3. 简单随机游走（SRW）和布朗运动（续）

中心极限定理

令 $\bar{X} = \frac{1}{N}\sum_{i=1}^{N} X_i$ 是随机样本的均值 $X_1, .., X_n$ 来自一个具有有限均值 μ 和有限正方差的分布，则

$$W = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \to \mathrm{N}(0,1), \text{ as } n \to \infty.$$

我们不需要知道 X_i 的底层分布，当 n 足够大时，这和正态样本（即 $X_1 \sim \mathrm{N}(\mu, \sigma^2)$））完全一样。